Salient Is Not Enough: Keeping Every Time Anchor for Temporal Grounding under a Visual-Token Budget

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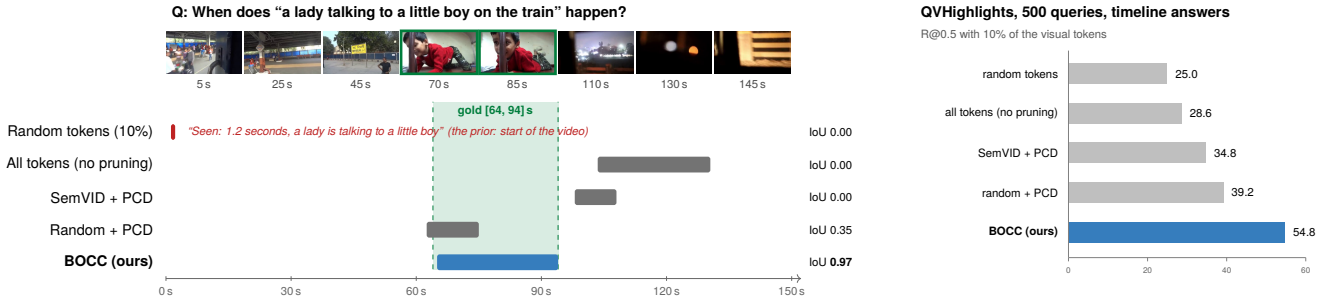


Figure 1. **Temporal grounding from a pruned video memory.** Left: a QVHighlights query, with the intervals predicted by each method at 10% of the visual tokens (real outputs). With random pruning the model falls back on a video-free prior and answers "1.2 seconds"; QA-oriented pruning and the uncompressed model pick a later, wrong scene. BOCC keeps every frame anchored in time and contrasts the answer against the model's no-video prediction, and recovers the interval (IoU 0.97). Right: R@0.5 over 500 queries when every method answers with a timeline of observations, the format of grounded QA. In this format the prior writes many of the times and BOCC is far above the uncompressed model. With a one-line answer the uncompressed model is much stronger, and Coverage alone matches it with a quarter of the tokens (Sec. 6.2).

Abstract

Video LLMs are increasingly asked not only what *happened but* when. *A grounded question such as "how long did I use the paper cutter?" needs both an answer and the time interval of its evidence. To keep long videos tractable, the visual memory of these models is pruned to a small fraction of its tokens, and published pruners lose little multiple-choice accuracy at 10% of the tokens. We show that temporal grounding needs something different:* pruning for QA keeps what is salient, but grounding needs what is everywhere. *A video LLM reads time from timestamps, and a timestamp survives only if some visual token of its frame survives. Pruners that select by saliency, uniqueness or text similarity empty whole frames, delete their timestamps, and fall below random token selection on grounding. When the answer lists observed moments before giving the interval, a video-free prior fills the gaps: asked with no video at all, the model still starts its intervals at the beginning of the window. We propose BOCC, a training-free method that selects frames by the question (BOLT) and adds two parts.* Coverage *keeps, in every layer, the tokens that the actual question attends to most, with an equal per-frame quota so that every time anchor survives. In the settings we tested, a global top-k on the same question attention does as well without the quota, so the question score is what drives the gain. Prior-contrastive decoding (PCD) writes each time value against the model's own no-video prediction. Because the two parts behave differently across answer formats, we study the format directly. With the standard one-line grounding prompt on the full Charades-STA and QVHighlights sets of TimeLens-Bench, Coverage at 31% of the tokens stays within 0.3 mIoU of the uncompressed model, for both a general and a grounding-tuned model of the same family. It beats random selection by $+0.5$ to $+1.6$ and a lower-resolution input of the same budget by $+2.1$ to $+2.6$, while PCD has no effect on a 513-query subset. In reasoning-style grounded QA (S-EMBER), PCD is the part that matters. With 10% of the tokens, BOCC improves on the uncompressed model by $+11.2$ and $+10.7$ mIoU under the same timeline prompt, and by $+5.5$ under the benchmark's official direct-answer prompt, although an input restricted to the gold evidence still scores higher in accuracy, and on every metric for the grounding-tuned model. Asking for a timeline of observations instead of a single in-*

terval costs single-moment grounding about 30 mIoU, so the answer format is a major variable when evaluating compressed memories. We also report where the method helps less. On short, densely sampled videos Coverage gains little on grounding (+0.5 mIoU on Charades-STA; NExT-GQA), although on NExT-GQA it keeps the accuracy of the uncompressed model at 10% of the tokens while random selection loses about four points. PCD helps only where the answer format lets the prior choose the times, and it lowers the scored grounding where it disturbs that format.

1. Introduction

Ask a video assistant "how long did I use the paper cutter?" and a useful answer has two parts, a duration and the interval it comes from. Such *grounded* questions are central to egocentric assistants, video search and step-by-step guidance. What makes them useful also makes them hard. The model must know when things happened across the whole window, including the moments just before and after the event.

At the same time, video LLMs pay for every frame. Qwen3-VL [38] spends about 330 visual tokens per frame (660 per pair of frames), so 64 frames already cost about 21,000 tokens, and long or streaming video multiplies this. Pruning the visual tokens to a small budget has therefore become the default, and a fast-growing literature reports that 10–25% of the tokens are enough to preserve multiple-choice accuracy [14, 18, 27, 35, 41, 44]. In this paper we ask what pruning does to *temporal grounding*. Our answer comes down to one principle with two consequences: **pruning for QA keeps what is salient, but grounding needs what is everywhere.** A QA answer can be read off a few informative frames. An interval can only be read from a timeline, and a timeline needs an anchor in every part of the window.

Observation 1: QA pruners lose the anchors. Video LLMs such as Qwen3-VL report time by reading timestamp tokens interleaved with the visual tokens. When all visual tokens of a frame are removed, its timestamp goes too, or it stays with nothing to show. Pruners designed for QA drop frames that look redundant, unremarkable or unrelated to the words of the question, and those are exactly the frames that make a timeline usable. On S-EMBER grounded QA, all four recent pruners we evaluate fall below random token selection at the same budget, and the same holds on QVHighlights and Charades-STA (Sec. 6.3). Random selection is a strong grounding baseline because it leaves a few tokens on every frame. In fact, one of the published scores improves by 5.3 mIoU just by ranking tokens within each frame instead of globally (Sec. 3.2).

Observation 2: where the memory is silent, a prior answers. Grounded QA invites a reasoning-style answer in which the model lists what it saw and when, and then states the interval. If we ask this way with no video at all, the model still produces confident intervals, and 90% of them start in the first 5% of the window. Pruned memories show the same habit: 30–56% of their predicted intervals start in the first 5% of the window, compared with 16% of the gold intervals, and removing the frames that contain the evidence raises this share from 37% to 51% (Sec. 3.3). So the prior does not belong to any particular pruner. It is what the decoder writes when the memory has nothing to say.

Observation 3: the answer format decides which failure dominates. These two observations call for two remedies, and the remedies do not matter equally everywhere. On the same QVHighlights queries and with the same memory, asking for a timeline of observations instead of a single interval costs about 30 mIoU (Sec. 6.2). With the one-line format, selecting tokens by the question helps and contrastive decoding does nothing. With the timeline format, the video-free prior writes many of the times, so contrasting it away helps, as long as the contrast does not break the answer format. For this reason we treat the answer format as an experimental variable instead of a fixed detail of the protocol.

We propose BOCC (BOLT + Coverage + Contrastive decoding), a training-free method with one component for each failure. An optional BOLT front end chooses which frames to encode (Fig. 3).

- **Coverage** (Sec. 4.1) keeps the anchors. Once the frames are encoded, we prune the KV cache layer by layer. Each frame gets an equal share of the layer's budget, and within each frame we keep the tokens that the actual question attends to most. A timestamp survives exactly when one of its frame's visual tokens survives, so the equal quota guarantees every time anchor while the question decides what each anchor shows. In the settings we tested, a global top-k over the same question attention does as well without the quota. The question score is therefore what drives the gain, and the quota acts as a safeguard for cases where scores collapse onto a few frames.
- **Prior-contrastive decoding (PCD)** (Sec. 4.2) removes the prior where it writes the answer. We choose each time value by contrasting the model's prediction with its prediction from a video-free memory. This is the content-free contrast used in contextual calibration and context-aware decoding [42, 52], applied only to time values and with a strength that fades where the model is already confident.

Neither operator is new on its own. Per-frame budgets appear in VidCom^2, KTV and AdaTP [1, 12, 35], question-attention KV eviction in SnapKV and PruneVid [27, 32], and contrast against a content-free or degraded input in contrastive decoding, classifier-free guidance and VCD [19, 30, 31, 40, 42]. What we contribute is an account of *why* temporal grounding needs each of them and when each one matters, together with a way to make them work together:

1. **A diagnosis of grounding under compression** (Sec. 3). QA-oriented pruners empty frames and fall below random selection. In reasoning-style answers the decoder's video-free prior fills the gaps, and an evidence-removal test shows this causally.
2. **Coverage**, per-layer question-driven token selection under an equal quota (Sec. 4.1). With the standard one-line prompt, on the full Charades-STA and QVHighlights sets and on both Qwen3-VL-8B and the grounding-tuned TimeLens-8B, it matches the uncompressed model with 24% of the visual tokens. It also beats random selection ($+0.5$ to $+1.6$ mIoU) and a matched low-resolution input ($+2.1$ to $+2.6$) (Sec. 6.2). On NExT-GQA at 10% of the tokens it keeps the accuracy of the uncompressed model, while random selection loses about four points ($+3.4$ and $+3.2$ over random). With BOLT frames it exceeds a 1 fps input of similar budget by $+9.2$ and $+6.5$ mIoU. Random tokens on the same frames, however, already gain $+7.0$ and $+6.1$, so most of that gain comes from the front end.
3. **PCD and an answer-format study** (Secs. 4.2 and 6.2). On S-EMBER, PCD helps reasoning-style answers on all eight memories we test and on both models. With 10% of the tokens, BOCC improves on the uncompressed model by $+11.2$ and $+10.7$ mIoU under the same prompt. Under the official S-EMBER prompt the grounding gain is $+5.5$, while the accuracy gain is smaller ($+2.9$) and comes mostly from BOLT. The same study shows that the answer format alone moves single-moment grounding by about 30 mIoU and decides whether contrastive decoding helps. PCD does nothing for one-line answers, and on NExT-GQA it lowers the scored grounding where it disturbs the format (TimeLens-8B timeline answers, minutes:seconds under the official prompt).
4. **Reproducible evaluation** (Secs. 5 and 6.2). We reproduce the published TimeLens-Bench numbers within 0.3 mIoU and run sanity checks with no video and with shuffled frames or timestamps. We also report a broad set of negative results, including the limits of Coverage on short, densely sampled videos and of PCD outside the timeline format.

2. Related Work

Frame selection. BOLT [34] scores frames against the query with a contrastive image–text model and then picks them by inverse transform sampling. AKS [43], MDP3 [5], FOCUS [2] and Q-Frame [6] optimise relevance jointly with coverage or diversity. All of them are evaluated on multiple-choice QA. Our frame selection front end (Sec. 4.3) adapts BOLT to grounding, and we find that it adds little once the tokens are chosen by the question.

Visual token pruning for video LLMs. Query-agnostic methods compress the video by redundancy or saliency. FastVID [41] segments the video and keeps density peaks, VidCom^2 [35] sets each frame's budget by how unique the frame is, and FlashVID [18] selects tokens by [CLS] attention and merges them across frames with trees. DyCoke [44] and FrameFusion [21] merge similar tokens. Query-aware methods also use the text. SemVID [14] scores tokens by their word-embedding relevance with MMR diversity, AdaTP [1] boosts segments that resemble the text, LGTTP [4] reallocates tokens according to temporal words in the question, and PruneVid [27] keeps the top 40% of tokens by question attention at a single LLM layer. KTV [12] is the closest in structure, since it selects keyframes first and then tokens under a relevance-weighted per-frame budget. LGTTP is the only method we know of that evaluates grounding, and it does so at a milder budget and offline. BOCC differs from these methods in three ways. Its per-frame quota is equal by design, its score is the attention of the actual question propagated exactly through all layers, and the quota is there to keep every timestamp. We show that emptying frames hurts grounding. We also show that once the actual question's attention is used, the quota acts as a safeguard and is not where the gain comes from (Sec. 6.4).

KV-cache compression and video memory. In long-context LLMs, SnapKV [32], H_2O [51], PyramidKV [15] and Ada-KV [20] evict KV entries by attention. Our token stage is SnapKV-style eviction applied to video, with a per-frame constraint added. For streaming video, HERMES [10] compresses the cache with generic pseudo-queries, ReKV [17] retrieves KV blocks when the question arrives, and StreamMem [8] and InfiniPot-V [3] maintain bounded memories. These works report QA only. We build our memory for each question from the frames seen up to the question time. It is therefore causal, but it is not a persistent streaming memory, and we make no claim about the cost of building it (Sec. 7).

Contrastive decoding and prior correction. Contrasting a model's prediction with that of a weaker or less informed model is an established decoding tool. Contrastive Decoding [31] subtracts a small "amateur" model and introduced the plausibility constraint that we use. Classifier-free guidance for language models [40] and context-aware decoding [42] amplify the difference between the predictions made with and without the conditioning context, which is the form of our Eq. (4). The idea of measuring the prior with a content-free input and correcting for it is older still. Contextual calibration [52] rescales label probabilities using the model's answer to "N/A", and domain-conditional PMI [24] reweighs answers by their task-conditioned prior. For vision-language models, VCD [30] contrasts against a noised image, ICD [46] against a disturbed instruction and M3ID [19] against the absence of the image, while PAI [33] amplifies image attention and subtracts text-only logits. SEASON [7] carries the idea over to video with tem-

porally corrupted negatives. These methods target hallucination, faithfulness or classification. PCD uses the same contrast for another purpose. We apply it only to *time values*, with a strength that adapts to the model's confidence, and we study when it helps and when it does not (Secs. 6.2 and 6.5).

Grounding-tuned video LLMs. TimeChat [39], VTimeLLM [25] and LITA [26] train video LLMs to localise moments using timestamp-aware encoders, boundary-aware training stages or dedicated time tokens. TimeLens [50] re-annotates the standard grounding benchmarks and fine-tunes Qwen-VL models with reinforcement learning. All of these works improve grounding with a full visual memory. We use TimeLens-8B as our grounding-tuned model and TimeLens-Bench as the standard protocol, and we ask what happens to such a model when its memory is pruned.

Training-free temporal grounding. NumPro [48] draws frame numbers onto the frames, FV-Action [9] scans the video with yes/no questions, and REZE [13] recasts grounding as recognition. [Authors to verify] [11] read intervals from calibrated attention heads and debias them against a run on a blank video. All of these methods assume the full set of video tokens. BOCC addresses the complementary problem of grounding from a *compressed* memory, and these readouts could run on top of it **[TODO: run FV-Action and the attention readout on the same 10% memory]**.

Benchmark biases. Otani et al. [37] showed that blind baselines are competitive on Charades-STA and ActivityNet Captions because the annotations carry positional priors.

3. Grounding under a Token Budget: a Diagnosis

We first establish Observations 1 and 2 of the introduction on S-EMBER, before introducing any method. Observation 3, which concerns the answer format, is studied in Sec. 6.2. Unless stated otherwise, the reference memory in this section is random 10% pruning.

3.1. Setting

A question x is asked at time q about a video $v_{0:q}$. The model sees F=64 frames from $[0, q]$, which are encoded into a KV cache $\mathcal{M}$ of N visual tokens. Timestamp tokens are interleaved with the visual tokens, and Qwen3-VL uses one "`<t seconds>`" for each pair of frames. A budget ρ (by default ρ=0.1) limits the number of tokens kept in each layer to ρN, and kept timestamps count toward this budget. The model answers with a short *timeline*:

Table 1. **Where pruning keeps tokens.** Share of the visual tokens kept, by position in the window (S-EMBER, 10% budget), and the resulting grounding (mIoU) and leak. FastVID (mIoU 17.9, leak 38%) keeps 6.0 / 8.5 / 10.5 / 10.6 / 12.9% over the five fifths of the window, and tokens from only 25% of the frames.

Selector	0–10%	10–25%	25–50%	50–75%	75–100%	mIoU	Leak
HERMES	4.7	3.9	2.6	2.5	27.6	18.3	49%
Stratified	8.7	8.7	8.6	8.6	8.7	23.7	**29%**
Random	8.9	8.8	8.6	8.7	8.6	**25.8**	30%

```
Seen: t_1 seconds, …
Seen: t_k seconds, …
Answer: …
Time: [s, e]
```

and we parse the interval $[s, e]$ from the last line. This reasoning-style format suits grounded QA, where the answer describes what was seen before it gives the interval, and we use it throughout the diagnosis. Section 6.2 compares it with the one-line format of grounding benchmarks (*"The event happens in s–e seconds"*). Grounding is scored by IoU with the gold interval, reported as mIoU and R@$\{0.3, 0.5, 0.7\}$. Grounded QA also requires the answer to be judged correct, so GQ@0.5 counts answers that are correct and have IoU $\geq$ 0.5. We call an answer an *early start* when its interval starts in the first 5% of the window, and a *leak* when it starts early even though the gold evidence starts later.

3.2. QA pruners lose temporal coverage

At 10% of the tokens, FastVID, VidCom^2, FlashVID and SemVID all score below random token selection (Sec. 6.3). Where the kept tokens lie explains why (Tab. 1). HERMES scores tokens by attention plus recency, so it keeps 27.6% of the tokens in the last quarter of the window and only 2.5–4.7% elsewhere. FastVID retains one representative per run of similar frames, and as a result it keeps a token from only 25% of the sampled frames in a median question. Those runs of similar frames are exactly where a continuing event lives. Random selection keeps about 9% everywhere and never empties a frame. Because a timestamp is useful only when some visual content of its frame survives with it, a selector that empties frames leaves holes in the timeline. These holes hurt even when the score itself is sensible. If we rank HERMES's scores within each frame instead of globally, which gives every frame an equal share ("stratified"), mIoU improves by 5.3.

3.3. The prior fills the gaps

The model answers without video. When we ask every question with an empty visual memory, the model still writes a timeline. Of its first `Seen:` moments, 89% are at 0 s, and 90% of its intervals start in the first 5% of the window, which gives an 88% leak. The median interval is 0.45× the gold

length, and 87% of the ends fall more than half an interval too early. We call this habit of starting early and stopping soon the model's *temporal prior*.

The prior leaks into answers with video. The same signature appears, more weakly, in every memory (Fig. 2). About 30% of intervals start early at every random budget and also with no pruning at all, compared with 16% of gold intervals. The share rises when coverage is lost, to 49% for HERMES and 37–39% for HERMES in streaming mode. The leak therefore depends on where the memory is silent and not on how many tokens it keeps.

Removing the evidence causes the leak. To test causality, we drop the uniform frames that fall inside the gold interval ("evidence"), or the same number of frames elsewhere ("control"), and keep every other frame with its true timestamp (Tab. 2). Without the evidence, the answers do not spread out. They move to the start of the video instead, and the leak rises by 14 points for both the unpruned and the random 10% memory. In the HERMES 10% memory, the leak also tracks how much of the evidence survives. Questions whose evidence keeps under 5% of its tokens leak 56% of the time, compared with 33% when at least 15% survives.

An instruction does not remove it. We added "The relevant moments may occur at any point in the video; do not assume they are near the start" to the prompt. The leak barely moved (29.5% → 28.0%), and grounding did not change (−0.1 mIoU). In this answer format, then, the prior acts in the output distribution and has to be removed there. We do not yet know whether the same prior appears when the model answers with a single interval. In that format PCD has no effect (Sec. 6.2). Those runs omit the answer reserve, which on its own changes S-EMBER grounding by only +0.1. The null effect suggests that the prior decides fewer of the written times there. **[TODO: no-video answers and leak under the one-line and direct-answer prompts.]**

The end is the weak boundary. Without contrastive decoding, the median end error across memories is 0.40–0.69 of the gold length, while the median start error is 0.14–0.30. The most frequent failure is an interval that lies entirely *before* the gold one, which accounts for 33% of all answers (Tab. 26).

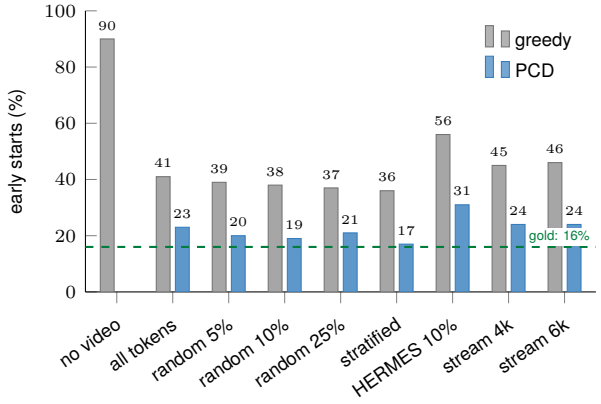


Figure 2. **The temporal prior leaks into every memory, and PCD removes most of it.** Share of predicted intervals that start in the first 5% of the window (S-EMBER). Without video the model starts early 90% of the time; every memory inherits part of this habit, most where pruning loses coverage (HERMES). PCD roughly halves it everywhere.

Table 2. **Causal test.** Frames inside the gold evidence ("evidence") or as many frames elsewhere ("control") are removed. Acc. / mIoU / GQ@0.5 and leak; leak measured on the 397 questions whose evidence starts after the first 5% of the window. PCD: Sec. 4.2.

		Greedy		PCD	
Memory	Removed	Acc./mIoU/GQ	Leak	Acc./mIoU/GQ	Leak
Random 10%	control	16.6/28.8/8.8	**31%**	15.4/34.6/10.5	**11%**
	evidence	7.8/ 8.6/1.7	46%	9.3/10.3/1.1	23%
All tokens	control	**19.6/29.4/10.1**	37%	**18.1/35.9/12.6**	15%
	evidence	9.1/ 9.4/1.9	51%	9.1/10.5/1.5	22%

4. Method: BOCC

BOCC intervenes at the two places where Sec. 3 found the failures. Coverage acts on the memory and prior-contrastive decoding (PCD) acts on the decoder, and both sit behind a frame-selection front end (BOLT). No component needs training, and Fig. 3 gives an overview.

4.1. Coverage: equal quotas filled by the question

Once the F frames are encoded, we prune the memory a single time, separately in each of the L layers. Let $\mathcal{T}_f$ be the visual tokens of frame pair $f \in \{1, \dots, F'\}$ and B the per-layer budget for visual tokens.

Question relevance. We append the question x to the full, uncompressed memory and run it exactly through all layers, so the queries at layer ℓ come from the real hidden states. A visual token i is scored at layer ℓ by the attention it receives from the question, averaged over heads h and question positions j:

$$a_\ell(i) \;=\; \frac{1}{H\,|x|} \sum_{h=1}^{H} \sum_{j \in x} \mathrm{softmax}_i\Big(\tfrac{1}{\sqrt{d}}\, Q_j^{(\ell,h)}\, K_i^{(\ell,h)^\top} \Big). \tag{1}$$

Equal quota. Each frame pair gets the same share of the budget, $B_f = \lfloor B/F' \rfloor$, with any remainder handed out in order. Layer ℓ then keeps

$$\mathcal{K}_\ell \;=\; \bigcup_{f=1}^{F'} \mathrm{top}_{B_f} \{\, a_\ell(i) \;:\; i \in \mathcal{T}_f \,\}. \tag{2}$$

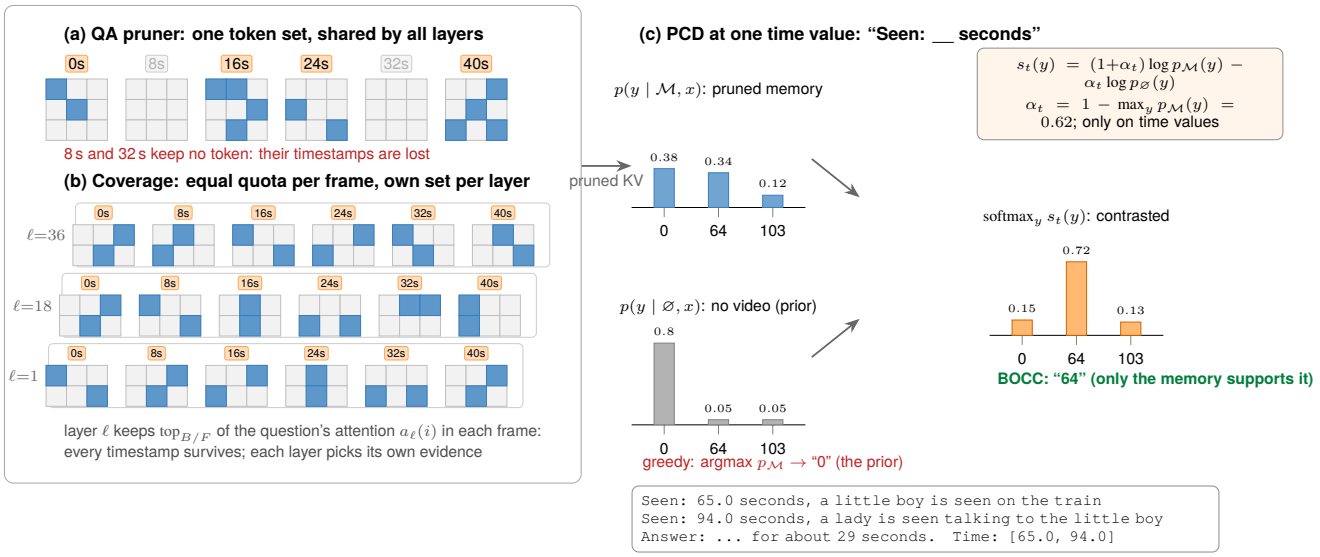


Figure 3. **Overview of BOCC. (a)** QA-oriented pruners keep one set of salient or unique tokens in every layer and can empty whole frames, removing their timestamps. **(b)** Coverage prunes each layer separately: every frame keeps the same share of the layer budget, filled with the tokens that the question attends to most *in that layer*. Every timestamp survives in every layer, and different layers keep different evidence (sharing one set across layers costs 1.5 mIoU, Tab. 15). **(c)** PCD at one time value of the teaser example (probabilities illustrative). Greedy decoding picks "0", which the memory supports only as much as the video-free prior does; the contrasted score discounts it and promotes "64", which only the memory supports. Output shown is the real BOCC answer.

The timestamp tokens of f are kept if and only if $\mathcal{K}_\ell \cap \mathcal{T}_f \neq \emptyset$. At any practical budget the equal quota leaves every frame pair with at least one token, so every time anchor survives, while the question decides which part of each frame is shown. Kept tokens keep their original rotary positions, which preserves their temporal order.

Why equal, and why the question. We chose the equal quota as a deliberately simple coverage constraint. It guarantees the timeline, which grounding needs and multiple-choice QA does not. It also keeps attention from collapsing onto a few frames. When quotas follow attention mass instead, attention sinks and recency bias can give a single frame up to 227 of the roughly 2,140 kept tokens. In a pre-registered test, relevance-weighted quotas with a floor, computed from SigLIP similarity or from attention mass, did not beat equal quotas (-0.2 mIoU). Attention becomes informative once it comes from the *actual* question. Streaming memories such as HERMES [10] score tokens with generic probe questions instead, and generic probes with a global top-k are the weakest variant we measured (18.3 mIoU); adding the equal quota lifts them by 5.3. With the actual question, however, a global top-k without any quota has so far done as well as the equal quota (Sec. 6.4). In that setting the quota acts as a guarantee and is not where the gain comes from.

Per-layer selection. Each layer keeps its own set $\mathcal{K}_\ell$. If we force a single shared set by averaging attention over layers, BOCC changes by -1.5 mIoU, while random selection is unaffected ($+0.7$). Per-layer selection therefore helps question-driven scores specifically, and it does not explain why random selection beats input-level pruners.

Frames and budget. An equal quota only works if each frame gets enough tokens to show something. With 2–4 tokens per frame pair, the most-attended tokens carry little content and Coverage collapses (Sec. 6.7). For small budgets we therefore spend the tokens on fewer frames. At a fixed number of kept tokens, 32 frames with twice the share per frame recover what 64 frames lose. Our rule is to use $F{=}64$ when each frame pair keeps at least $\sim$8 tokens and to halve F otherwise. Kept timestamps count against the budget, and at budgets of a few hundred tokens they take up a large part of it, so some of the collapse may come from this accounting (Sec. 6.6). **[TODO: this rule was derived from the results of Sec. 6.7; validate it on held-out data and with the fixed timestamp rule.]**

Cost. For each question, Coverage adds one forward pass of the question over the uncompressed memory. Because the full memory is prefilled first, Coverage shrinks the memory that the decoder reads but does not lower the cost of building it. PCD decodes a second, prompt-only row in lockstep at every step, even though its scores are used only inside time values. This roughly doubles the decoder's per-step work, on a much shorter context. **[TODO: measured prefill, time to first token, total latency, peak memory and KV size; the current implementation runs 3–4 scoring passes, reducible to one.]**

4.2. PCD: prior-contrastive decoding of time values

Let $p_{\mathcal{M}}(y) = p(y \mid \mathcal{M}, x, y_{<t})$ be the next-token distribution with the compressed memory and $p_{\varnothing}(y) = p(y \mid \varnothing, x, y_{<t})$ the distribution with the same prompt and generated prefix but no visual tokens at all. Then

$$\log p_{\mathcal{M}}(y) = \log p_{\varnothing}(y) + \underbrace{\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}}_{\text{evidence from } \mathcal{M}}, \tag{3}$$

so when the memory says nothing about y, the evidence term is flat and the video-free prior decides. This is the failure described in Sec. 3.3. Prior-contrastive decoding (PCD) amplifies the evidence term,

$$\begin{aligned} s_t(y) &= (1+\alpha_t)\log p_{\mathcal{M}}(y) - \alpha_t \log p_{\varnothing}(y) \\ &= \log p_{\varnothing}(y) + (1+\alpha_t)\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}, \end{aligned} \tag{4}$$

and picks $y_t = \arg\max_{y \in \mathcal{V}_\beta} s_t(y)$ over the plausible set $\mathcal{V}_\beta = \{y : p_{\mathcal{M}}(y) \geq \beta \max_{y'} p_{\mathcal{M}}(y')\}$ with β=0.1, as in VCD [30], so that tokens the memory makes implausible are never promoted. We adapt this rule to grounding in three ways.

No-video counterfactual. We use an empty negative memory instead of a noised or shuffled one. This isolates exactly the prior measured in Sec. 3.3, and it is cheap, since its cache holds only the prompt.

Time scope. PCD is active only while the model writes a time value, that is, inside a `Seen:` time before its description and inside `Time: [s, e]`. Every other token, including the answer text, is decoded greedily from $p_{\mathcal{M}}$, so PCD leaves the answer alone wherever the memory already provides it.

Confidence-adaptive strength. We set $\alpha_t = \alpha_{\max}\big(1 - \max_y p_{\mathcal{M}}(y)\big)$ with $\alpha_{\max}$=1. When the memory makes the model confident, *e.g.* because the evidence really starts at 0 s, PCD fades out and the early answer stands. When the model hesitates between several times, the evidence term decides.

The prompt format matters here. Under the timeline prompt, PCD first acts on the listed `Seen:` moments and then on the interval, so the interval is written from observations that have already been contrasted. With PCD we also reserve 64 tokens of the generation budget for the final `Answer:` and `Time:` lines: once 64 tokens remain and no `Answer:` line has started, the decoder begins one. By itself the reserve changes S-EMBER mIoU by only +0.1 and repairs only three truncated answers. When the answer is a single interval, PCD has little to correct. With the one-line grounding prompt it changes mIoU by −0.1 (Sec. 6.2), in runs without the reserve. For the official direct-answer prompt we do not yet have a clean PCD-only comparison on S-EMBER. On NExT-GQA, PCD under that prompt lowers the scored grounding, because more times are written as minutes:seconds (Sec. 6.5).

4.3. BOLT: question-guided frames

When the window is much longer than F frames, the choice of frames to encode can matter as much as the choice of tokens to keep. Our front end samples frames as in BOLT [34]. We embed each second of video once with SigLIP 2 [45], which gives a causal 1 fps index. The relevance z_j of second $j \leq q$ is its z-scored cosine similarity with the question, and the F frames are taken at evenly spaced quantiles of

$$p(j) = \tfrac{1}{2} \cdot \tfrac{1}{q} + \tfrac{1}{2}\,\mathrm{softmax}(z)_j, \tag{5}$$

so half of the frames remain uniform. We skip BOLT when uniform frames would be less than 1 s apart. Once the question chooses the tokens, BOLT adds only +0.8 to +0.9 mIoU on S-EMBER, QVHighlights and ActivityNet. The runs in the main tables include it.

5. Experimental Setup

Models. Our main model is Qwen3-VL-8B-Instruct [38]. It sees 64 frames at native resolution, sampled uniformly from $[0, q]$ or chosen by BOLT (Sec. 4.3), and decodes greedily or with PCD, at batch size 8 for pruned memories. On S-EMBER we generate at most 384 new tokens with repetition penalty 1.1. On TimeLens-Bench the limit is 512, and on NExT-GQA it is 256 for the two-query setting or 1,024 for the joint answer, both with repetition penalty 1.0. Unless stated otherwise, all methods share the timeline prompt. The exceptions are the full TimeLens-Bench sets, including the QVHighlights and Charades-STA columns of Tab. 3, and the prompt study of Sec. 6.2, which use the one-line prompt of TimeLens-Bench. In Sec. 6.2 we also add TimeLens-8B [50], a Qwen3-VL-8B model fine-tuned for grounding. TimeLens-8B frames are capped at about 1.57 MP on S-EMBER and NExT-GQA. On TimeLens-Bench both models use the protocol input (2 fps, at most 14,336 visual tokens and at least 64 per frame), and BOLT runs there use 64 or 128 frames of at most 0.79 MP. Batch size alone changes about half of the greedy outputs and moves a score by up to about two points. Comparisons with the uncompressed model, which runs at batch size 1–2, include this noise.

Two clusters. We made the later runs on a second GPU cluster. These include most fixed-rule and all 5% NExT-GQA runs, BOLT on the full QVHighlights set, the official S-EMBER prompt, and the TimeLens-8B gold-informed runs. Some paired differences combine runs from both clusters (marked where they occur). Runs repeated on both clusters agree within one mIoU point. NExT-GQA without pruning moves by +0.4 mIoU, with 469 of 510 replies

identical. On S-EMBER, random tokens move by $+0.2$ and BOCC by -0.9 mIoU, and the TimeLens script on QVHighlights moves by -0.2.

Datasets. **S-EMBER** grounded QA [47]: 475 questions (191 duration, 184 counting, 100 location) on the first 300 videos, asked mid-stream, so the memory contains only the frames before the question. **QVHighlights** [29] val: 500 queries with a single relevant window, sampled with seed 0 from the 1,020 single-window queries, on videos of about 150 s. **ActivityNet Captions** [28] val_1: 500 queries, one sentence per video, on videos of about 115 s. **Charades-STA** [22] test: 500 queries on videos of about 30 s. For these three datasets the question is *"When does the following happen in the video: <query>?"* and the window is the whole video. **TimeLens-Bench** [50]: the re-annotated Charades-STA, QVHighlights and ActivityNet sets, evaluated with their one-line prompt on the full sets and on a 513-query QVHighlights subset. **NExT-GQA** [49] test: 1,005 questions on videos of about 40 s. The answer and the interval are asked either in two separate queries (multiple choice or open-ended) or, for open-ended answers, in one joint reply. Our QVHighlights and ActivityNet copies are low-resolution re-encodes with 153 and about 84 visual tokens per frame pair, compared with 660 on S-EMBER, so we also report budgets in absolute tokens.

Budget. The default budget is 10% of the visual tokens of the 64-frame memory, timestamps included (about 2,140 tokens on S-EMBER, about 520 on QVHighlights). We report ActivityNet at 30% (400–900 tokens), because at 10% its memory holds only 120–300 tokens (Sec. 6.6). Since timestamps count against the budget, a KV selector at a nominal ρ keeps fewer visual tokens than an input pruner at the same ρ; on NExT-GQA at 10% it keeps about 3.4% of the visual tokens. A frame's timestamp is kept when the frame keeps a visual token. An earlier version of this rule could let timestamps fill the budget of very small memories and leave no visual token. The NExT-GQA 10% runs use the fixed rule (Sec. 6.2) and the other runs use the earlier one. The difference can matter only where timestamps take a large share of the budget (Sec. 6.6).

Baselines. We compare against all tokens (no pruning), random tokens, and four QA pruners: FastVID [41], VidCom^2 [35], FlashVID [18] (vision-side stages) and SemVID [14]. The pruners are re-implemented and checked to keep the same tokens as the authors' code on test inputs. Each baseline is run with and without PCD. A global question top-k without quotas is compared in Sec. 6.4. **[TODO: SnapKV-style observation-window eviction, PruneVid, and FV-Action / attention readout on the same memory.]**

Metrics. We report mIoU and R@$\{0.3, 0.5, 0.7\}$. On S-EMBER we also report answer accuracy and GQ@0.5, judged by a local Qwen3.8-27B-FP8 model with the official S-EMBER judge prompt. Judging was repeated in three sessions, whose verdicts for the same run differ by up to about one accuracy point. We therefore never pair Acc. or GQ@0.5 across sessions, and each table uses a single session, named in its caption: session A for Tab. 4, B for Tabs. 9 and 23, and C for Tab. 24. The early-phase tables listed in Sec. 6 use their own earlier sessions and are compared only within themselves. The same Qwen3-VL-8B BOCC run scores 21.7 accuracy and 13.3 GQ@0.5 in session A and 21.9 and 13.5 in session B. The TimeLens-8B BOCC run scores 27.4 and 17.5 in B and 26.5 and 17.1 in C. Open-ended NExT-GQA answers are judged by the same model with the Video-ChatGPT prompt [36], in separate batches that share a cache. On S-EMBER, answers that give times as minutes:seconds are converted to seconds, for every method alike (Appendix A). The joint NExT-GQA answers are scored with the official S-EMBER parser, which counts such times as missing.

Protocol. We made all design choices (scores, quotas, α, scope) on S-EMBER. We then froze the configuration and ran it once on QVHighlights, ActivityNet and Charades-STA. **[TODO: held-out S-EMBER videos as a test split.]**

6. Results

We organise the results by question. Section 6.1 compares BOCC with all baselines on three datasets, each with the prompt that gives the methods their best results there: the reasoning-style timeline prompt on S-EMBER and the standard one-line prompt on QVHighlights and Charades-STA. Section 6.2 then studies the answer format, a grounding-tuned model and NExT-GQA. Section 6.3 looks at how published pruners behave on grounding, and Secs. 6.4 and 6.5 analyse the two components. Sections 6.6 and 6.7 study the budget and how to spend it. The remaining sections compare with upper bounds that use the gold interval (Sec. 6.8), analyse BOLT (Sec. 6.9), list what did not work (Sec. 6.10) and examine the errors that remain (Sec. 6.11).

Reading the tables. Numbers are mIoU, or mIoU / R@0.5 where two are given. Δ is a paired difference. Tables built from the early S-EMBER phases (Tabs. 16 to 19 and 26 and Tab. 2) use an earlier implementation of PCD, the official parser and batch size 1. On the S-EMBER rows we checked, the two parsers differ by 0.2–0.7 mIoU. The exception is the oracle selector with PCD, where they differ by 1.5 mIoU because 17 answers are written in minutes:seconds. All other S-EMBER tables use the minutes:seconds-aware parser. In every table the best value is in **bold** and the second best underlined, per column unless a caption says per row. Differences (Δ) are not ranked.

6.1. Main results

Table 3 leads to three conclusions.

Table 3. **Main results**, mIoU / R@0.5, with Qwen3-VL-8B and the prompt that gives the methods their best results on each dataset. S-EMBER: timeline prompt, 475 questions, 64 frames, 10% of the tokens. QVHighlights and Charades-STA: the one-line TimeLens-Bench prompt on the full sets (1,541 and 3,363 queries) at the protocol input, with 31% of all tokens including timestamps (about 24% of the visual tokens); in these two columns, all tokens is the TimeLens evaluation script. BOLT is skipped on Charades-STA, where uniform frames are already dense (–). The bottom block uses BOLT frames on the full QVHighlights set at other budgets and is not ranked: 64 or 128 frames, keeping 1,192 or 4,455 tokens per query including timestamps. **[TODO:]** cells are not yet run.

Method	S-EMBER timeline, 10%	QVHighlights one-line, 31%	Charades-STA one-line, 31%
All tokens (no pruning)	28.0 / 26.7	59.6 / **64.5**	48.5 / **54.0**
Random tokens	25.6 / 24.6	58.1 / 62.6	48.2 / 52.3
FastVID [41]	17.9 / 15.8	**[TODO:]**	**[TODO:]**
VidCom^2 [35]	20.4 / 18.7	**[TODO:]**	**[TODO:]**
FlashVID [18]	22.7 / 21.5	**[TODO:]**	**[TODO:]**
SemVID [14]	23.3 / 20.6	**[TODO:]**	**[TODO:]**
Coverage (uniform frames)	31.7 / 32.4	**59.7 / 64.5**	**48.6 / 54.0**
BOLT + Coverage	32.5 / 34.1	**[TODO:]**	–
BOLT + Coverage + PCD (BOCC)	**39.3 / 40.8**	**[TODO:]**	**[TODO:]**
BOLT 64 frames + random tokens	–	55.0 / 59.8	–
BOLT 64 frames + Coverage	–	57.2 / 62.9	–
BOLT 128 frames + random tokens	–	59.9 / 65.3	–
BOLT 128 frames + Coverage	–	60.5 / 66.1	–

(i) QA pruners lose to random selection, and Coverage does not. On S-EMBER, all four published pruners score below random tokens at the same budget (Sec. 6.3), while Coverage on uniform frames scores 31.7 mIoU against 25.6 for random tokens and 28.0 for all tokens. We have not yet run the pruners with the one-line prompt on the full QVHighlights and Charades-STA sets. Under the timeline prompt they fall below random selection on both datasets (Fig. 4).

(ii) With the one-line prompt, Coverage keeps the grounding of the uncompressed model on both full sets. With about 24% of the visual tokens it scores 59.7 / 64.5 on QVHighlights and 48.6 / 54.0 on Charades-STA, against 59.6 / 64.5 and 48.5 / 54.0 without pruning. Random selection at the same budget loses 1.6 mIoU and 2.0 R@0.5 on QVHighlights, and 0.3 mIoU and 1.7 R@0.5 on Charades-STA, so Coverage leads it by +1.6 and +0.5 mIoU. The margin is small at this budget, and it is largest on QVHighlights, whose videos are longer. With BOLT frames on QVHighlights (bottom block), Coverage leads random tokens by +2.2 mIoU at 1,192 kept tokens and by +0.6 at 4,455. At 1,192 tokens it stays 2.5 mIoU below the uncompressed model, and at 4,455 tokens it is 0.9 above it (Sec. 6.2).

(iii) On S-EMBER, where answers are reasoning-style, both parts of BOCC matter. BOLT adds 0.8 mIoU to Coverage and PCD adds another 6.8, so BOCC reaches 39.3 / 40.8. This is +11.2 mIoU above the uncompressed model answering in the same format. It is also +4.2 above random tokens with PCD (35.0, Tab. 4), so the gain does not come from PCD alone. Under the one-line prompt PCD has no effect (Tab. 6), and the BOCC cells of the one-line columns are not yet run. Timeline-prompt results on 500-query subsets of QVHighlights, ActivityNet and Charades-STA are in Tabs. 13, 21 and 22. On single-moment grounding that format costs the uncompressed model about 30 mIoU (Tab. 6).

Table 4. **Grounded QA on S-EMBER** (10%; all rows from judging session A, Sec. 5). ΔAcc. *vs*. random tokens.

Method	Acc.	GQ@0.5	mIoU	ΔAcc.
All tokens	19.8	8.8	28.0	+2.5
Random tokens	17.3	7.4	25.6	–
FastVID	15.2	4.8	17.9	−2.1
VidCom^2	16.2	5.9	20.4	−1.1
FlashVID	17.3	6.1	22.7	+0.0
SemVID	19.8	7.6	23.3	+2.5
Coverage, uniform frames	**22.5**	11.2	31.7	+5.3
Random + PCD	20.4	12.2	35.0	+3.2
SemVID + PCD°	17.9	9.9	32.3	+0.6
BOCC	21.7	13.3	**39.3**	+4.4
BOCC + coverage gate	22.1	**13.5**	**39.3**	+4.8

° earlier implementation of PCD.

Answers. BOCC also improves the answers (Tab. 4). Its accuracy is 21.7, against 17.3 for random tokens (+4.4) and 19.8 for all tokens. GQ@0.5 rises to 13.3, from 7.4 and 8.8 respectively. These numbers come from judging session A; in session B the same BOCC run scores 21.9 and 13.5 (Tabs. 9 and 23). Coverage alone gives the best accu-

Table 5. **S-EMBER by question type**, mIoU / R@0.5 (10%).

Method	Duration (191)	Counting (184)	Location (100)
All tokens	37.6 / 40.3	30.1 / 25.5	5.9 / 3.0
Random tokens	33.2 / 35.1	29.6 / 26.1	3.5 / 2.0
BOLT + Coverage	43.2 / 46.6	35.3 / 36.4	7.1 / 6.0
Random + PCD	41.0 / 40.8	41.9 / 40.2	**11.0 / 9.0**
SemVID + PCD	38.9 / 36.6	44.6 / **50.0**	8.9 / 8.0
BOCC	**47.6 / 50.8**	**46.9** / 48.4	9.3 / 8.0

racy (22.5), so PCD trades about one point of accuracy for its grounding gain. Differences below about two points of accuracy are within the decoding noise of the judge setup. **[TODO: SemVID + PCD in the same judging session.]**

Question types. BOCC improves duration and counting questions most (Tab. 5). Location questions ("where did I put …") remain essentially unsolved, with mIoU $\leq$ 11 for every method. Part of the reason is the labels. For 62 of the 100 location questions, the annotators disagree on the interval (mean pairwise IoU $<$ 0.6), which caps what any method can score.

6.2. Standard prompt, a grounding-tuned model, and short videos

Apart from the one-line columns of Tab. 3, the results so far use one model and the reasoning-style timeline prompt. Here we test the method in three further settings: under the standard one-line grounding prompt of TimeLens-Bench [50], on a grounding-tuned model (TimeLens-8B [50]), and on NExT-GQA [49]. Unless stated otherwise, paired differences are computed on the same questions. Unpruned runs use batch size 1–2 and pruned runs batch size 8, and this difference alone can move scores by up to about two points (Sec. 5). Some runs were made on a second GPU cluster. We flag every paired difference that combines runs from the two clusters; repeated runs across clusters differ by less than one mIoU point (Sec. 5).

Our pipeline reproduces published numbers. With the TimeLens evaluation script, Qwen3-VL-8B scores 48.5, 59.6 and 47.1 mIoU on the full Charades-STA, QVHighlights and ActivityNet sets of TimeLens-Bench, against published scores of 48.3, 59.4 and 46.8. TimeLens-8B scores 55.4, 65.7 and 53.2, against 55.2, 65.5 and 53.2. Our own pipeline without pruning also agrees with the script on the same queries (Tab. 27). The differences are +0.5 mIoU on Charades (400 queries), +1.4 on ActivityNet (400) and 0.0 on QVHighlights (513).

The timeline prompt is costly for single-moment grounding. Table 6 changes only the prompt. On QVHighlights queries, asking the model for a timeline of observations before the interval costs about 30 mIoU compared with the one-line prompt: −30.0 without pruning and −31.0 for

Table 6. **Answer format** (QVHighlights, 513 TimeLens-Bench queries, Qwen3-VL-8B, 300 frames, 31% of the tokens), mIoU. One-line: the TimeLens-Bench prompt. Timeline: the prompt of all other tables. The one-line PCD run used no answer reserve, the timeline PCD runs a reserve of 64 tokens (Sec. 4.2); on S-EMBER the reserve alone changes mIoU by +0.1.

	One-line prompt		Timeline prompt	
Selection	greedy	PCD	greedy	PCD
All tokens	60.5	–	30.4	–
Random tokens	58.2	–	**33.4**	43.9
Coverage	**61.7**	61.6	30.7	43.9

Coverage. PCD recovers part of this loss, adding +13.2 on Coverage and +10.4 on random tokens, yet it stays far below the one-line prompt (−17.8). Under the one-line prompt PCD has no effect (−0.1), and the same is true with BOLT (−0.3 and −0.5). The one-line PCD runs used no answer reserve, unlike the timeline runs, so this comparison is not matched on the reserve. We do not expect the reserve to explain the difference, because on S-EMBER the reserve alone does not change grounding (+0.1). **[TODO: one-line PCD with the 64-token reserve.]** With the timeline prompt, Coverage is not better than random selection on these queries (−2.8 greedy, +0.1 with PCD).

These results bear on the rest of the paper in two ways. The timeline-prompt results on QVHighlights, ActivityNet and Charades-STA (Fig. 1 and Tabs. 13, 21 and 22) are measured in a format that handicaps the uncompressed model, so the gains over "all tokens" seen there do not hold against the one-line prompt. For this reason the QVHighlights and Charades-STA columns of Tab. 3 use the one-line prompt. And PCD corrects time values in reasoning-style answers, where the model writes several times before the interval. When the answer is a single interval, there is little for it to correct.

Coverage under the standard prompt. With the one-line prompt (Tab. 7, top), Coverage at 24% of the visual tokens beats random selection on both models, by +3.5 and +2.0 mIoU. It also beats SemVID (+2.9) and the TimeLens input re-sampled to a similar budget (+3.7, +3.0). Against the uncompressed model it scores +1.2 mIoU for Qwen3-VL-8B and +0.2 for TimeLens-8B.

BOLT frames on the full QVHighlights set. On the full QVHighlights set (Tab. 7, bottom), at about 1.2k tokens per query, BOLT with Coverage exceeds a 1 fps TimeLens input capped at 1,365 video tokens by +9.2 mIoU on Qwen3-VL-8B and +6.5 on TimeLens-8B. BOLT with random tokens, however, already gains +7.0 and +6.1. Coverage adds +2.2 over random tokens for Qwen3-VL-8B and +0.4 for TimeLens-8B, and at 128 frames (about 4.5k tokens) it adds +0.6 and −0.0. Most of this gain therefore comes from

Table 7. **Standard grounding prompt** (QVHighlights, one-line prompt), mIoU / R@0.5. Top: 513 TimeLens-Bench queries, 300 frames at protocol size; Kept: visual tokens in memory (keep ratios count timestamp tokens too: 31% of all tokens is 24% of the visual tokens). Bottom: the full set (1,541 queries) with 64 or 128 frames chosen by BOLT; Kept: tokens per query *including* timestamps. TimeLens input: the TimeLens-Bench input at a lower resolution, capped at the given number of video tokens. Each block is ranked separately.

Method	Kept	Qwen3-VL-8B	TimeLens-8B
All tokens	13,650	60.5 / 64.9	67.1 / 73.1
TimeLens input, 2 fps low-res	4,200	58.0 / 62.0	64.3 / 68.8
Random tokens (31%)	3,271	58.2 / 60.8	65.3 / 70.4
SemVID (31%)	4,232	58.8 / 62.4	–
Coverage (31%)	3,271	**61.7 / 66.7**	**67.3 / 73.9**
All tokens, full set	all	59.6 / 64.5	65.7 / 71.8
TimeLens input, 1 fps low-res	≤1,365	48.0 / 51.5	57.0 / 61.4
BOLT 64 + random (23%)	1,192	55.0 / 59.8	63.1 / 69.7
BOLT 64 + Coverage (23%)	1,192	**57.2 / 62.9**	**63.5 / 70.0**
TimeLens input, 2 fps low-res	≤4,500	57.3 / 61.6	63.3 / 68.4
BOLT 128 + random (43%)	4,455	59.9 / 65.3	**65.3 / 71.1**
BOLT 128 + Coverage (43%)	4,455	**60.5 / 66.1**	**65.3** / 70.8

Table 8. **Full TimeLens-Bench sets** (one-line prompt, 31% of the tokens), mIoU / R@0.5. All tokens and low-res: the TimeLens evaluation script, the latter at a lower resolution with a cap of 3,500 / 4,500 *video* tokens on Charades-STA / QVHighlights. Coverage and random keep 3,348 / 4,652 tokens per query *including* timestamps, so the low-res input sees at least as many visual tokens. Paired ΔmIoU below.

	Qwen3-VL-8B		TimeLens-8B	
Method	Charades	QVHighlights	Charades	QVHighlights
All tokens (no pruning)	48.5 / **54.0**	59.6 / **64.5**	**55.4 / 63.1**	**65.7 / 71.8**
TimeLens input, low-res	46.0 / 49.6	57.3 / 61.6	52.8 / 59.6	63.3 / 68.4
Random tokens (31%)	48.2 / 52.3	58.1 / 62.6	54.5 / 61.7	64.2 / 69.6
Coverage (31%)	**48.6 / 54.0**	**59.7 / 64.5**	55.2 / **63.1**	65.4 / 71.0
Coverage *vs.* random	+0.5	+1.6	+0.7	+1.2
Coverage *vs.* low-res	+2.6	+2.4	+2.4	+2.1
Coverage *vs.* all tokens	+0.1	+0.1	−0.2	−0.2

Table 9. **A grounding-tuned model on S-EMBER** (10%, timeline prompt), mIoU / R@0.5 and Acc. / GQ@0.5. All Acc. and GQ@0.5 values are from judging session B (Sec. 5).

	Qwen3-VL-8B		TimeLens-8B	
Method	mIoU / R@0.5	Acc. / GQ	mIoU / R@0.5	Acc. / GQ
All tokens (no pruning)	28.0 / 26.7	19.6 / 8.8	33.1 / 34.1	20.6 / 12.2
Random tokens	25.6 / 24.6	17.1 / 7.2	28.2 / 26.7	19.6 / 9.5
Random + PCD	35.0 / 33.9	20.0 / 12.0	38.4 / 39.4	20.4 / 12.4
BOLT + Coverage	32.5 / 34.1	**22.3** / 11.2	37.2 / 37.9	22.9 / 13.1
BOCC	**39.3 / 40.8**	21.9 / **13.5**	**43.9 / 46.9**	**27.4 / 17.5**
BOCC *vs.* all tokens	+11.2	+2.3 / +4.6	+10.7	+6.7 / +5.3

choosing the frames by the question and keeping their resolution, not from Coverage. The comparison also leaves out the cost of the SigLIP index. On the 513-query subset at 897 visual tokens, BOLT + Coverage led a 1,125-token 1 fps input by +8.8 and +6.8, and Coverage led random tokens by +2.8 with Qwen3-VL-8B.

Both models lose grounding accuracy against their uncompressed memory at about 1.2k tokens (BOLT + Coverage: −2.5 and −2.1 mIoU). At about 4.5k tokens they are level with it (+0.9 and −0.4), and above the 2 fps low-resolution input capped at 4,500 tokens (+3.2 and +2.0). With the protocol input instead of BOLT frames, about 1.2k timestamp tokens use up a budget of 8.2%, so no visual token survives and both selectors collapse (4.6 and 4.2 mIoU). At small budgets, fewer frames are needed (Sec. 6.7).

Full TimeLens-Bench sets. On the complete Charades-STA (3,363 queries) and QVHighlights (1,541) sets of TimeLens-Bench (Tab. 8), Coverage at 31% of the tokens is lossless on both models, within −0.2 to +0.1 mIoU. It beats random selection in all four cells, by +0.5 to +1.6. It also beats, in all four cells, the TimeLens input reduced to the same budget by lowering the resolution (+2.1 to +2.6). Random selection, in turn, falls below the uncompressed model in all four cells, by 0.3 to 1.6 mIoU, and least for Qwen3-VL-8B on Charades-STA. The gap to random is largest on QVHighlights, where videos are longer. On Charades-STA, 31% leaves about 3,350 of 10,800 tokens, and random selection is already close. Keeping full resolution and choosing tokens works better than spending the same budget on a lower resolution. Even random selection beats the low-res input, by +2.2 and +1.7 on Charades-STA and by +0.8 and +0.9 on QVHighlights. **[TODO: full ActivityNet set (paused).]**

A grounding-tuned model on grounded QA. S-EMBER answers explain before they ground, so the timeline format is natural there. TimeLens-8B, which is trained for grounding, scores above Qwen3-VL-8B in every row of Tab. 9, and both models rank the methods in the same order. With 10% of the tokens, BOCC improves on the uncompressed TimeLens-8B on every metric: +10.7 mIoU, +6.7 accuracy and +5.3 GQ@0.5. Its lead over random selection is larger, at +15.7 mIoU, +7.8 accuracy and +8.0 GQ@0.5. Each component helps on its own. PCD adds +10.2 mIoU to random tokens and +6.7 to BOLT + Coverage, and BOLT + Coverage adds +9.0 to random tokens.

We repeated the judging in a second session (C), which also scores the gold-informed TimeLens-8B runs of Sec. 6.8. It gives BOCC 26.5 accuracy and 17.1 GQ@0.5, compared with 27.4 and 17.5 in session B. The two sessions agree within 1.1 accuracy points on every TimeLens-8B row, and every difference above keeps its sign; over all tokens, session C gives +6.7 accuracy and +5.5 GQ@0.5. Session C also shows that the grounding-tuned model prof-

Table 10. **NExT-GQA, multiple choice** (test, 1,005 questions, 64 frames, two-query protocol: one query for the answer, one for the interval; no PCD). Coverage and random budgets count timestamp tokens, so at 10% they keep about 3.4% of the visual tokens; SemVID, an input pruner, keeps the nominal share (matched comparison: Tab. 11). All tokens: batch size 2, first cluster; 10% and 5% rows: batch size 8, second cluster. † Earlier timestamp rule, which could leave no visual tokens on low-budget questions (about 282 of 2,010 prompts at 5%); reruns with the fixed rule are pending. All other KV-selector rows use the fixed rule or are unaffected (30%).

Model	Method	Acc@QA	Acc@GQA	mIoP	mIoU
Qwen3-VL-8B	All tokens	79.7	38.9	**48.0**	**37.7**
	Random 30%	78.6	38.6	47.5	37.3
	SemVID 30%	78.9	38.5	46.2	35.8
	Coverage 30%	**79.8**	**39.4**	47.9	37.6
	Random 10%	75.7	36.7	46.6	36.4
	Coverage 10%	79.1	38.5	47.4	37.0
	Random 5%†	68.3	27.3	40.5	30.5
	SemVID 5%	74.0	30.0	41.0	30.0
	Coverage 5%†	75.0	32.4	41.6	31.4
TimeLens-8B	All tokens	**76.7**	**41.5**	**52.1**	39.8
	Random 10%	72.8	38.6	50.1	38.9
	Coverage 10%	76.0	40.8	**52.1**	**40.0**
	Random 5%†	66.0	28.7	43.6	34.0
	Coverage 5%†	71.5	33.3	46.0	35.6

its more from the method. In it, with Qwen3-VL-8B rerun on the second cluster, BOCC on TimeLens-8B is above BOCC on Qwen3-VL-8B by +5.4 mIoU, +7.0 accuracy and +5.5 GQ@0.5. With random tokens the two models differ by +2.4 mIoU. TimeLens-8B's own gold-informed upper bounds, however, remain well above BOCC (Sec. 6.8).

Short videos. NExT-GQA videos last about 40 s, so 64 frames already sample them densely (Tab. 10). At 10%, Coverage is within 0.7 points of the uncompressed model on every metric for both models. For Qwen3-VL-8B the differences are −0.6 Acc@QA, −0.4 Acc@GQA and −0.7 mIoU; for TimeLens-8B they are −0.7, −0.7 and +0.2. Random selection, by contrast, loses about four points of accuracy (−4.0 and −3.9; these pairs combine the two clusters). Coverage therefore beats random selection on Acc@QA, by +3.4 with Qwen3-VL-8B and +3.2 with TimeLens-8B. On grounding the two selectors differ little. Coverage leads by +0.6 mIoU for Qwen3-VL-8B. For TimeLens-8B it leads by +1.9 mIoP, +1.1 mIoU and +2.2 Acc@GQA. At 30%, Coverage is lossless (+0.5 Acc@GQA, −0.2 mIoU), is +1.2 Acc@QA above random selection, and grounds better than SemVID (+1.8 mIoU). When frames are dense, then, where tokens are kept matters most for the answer and less for the interval, as on Charades-STA (Sec. 6.7).

The 5% rows still use the earlier timestamp rule. Under that rule, Coverage leads random selection on Acc@QA by +6.8 with Qwen3-VL-8B and +5.6 with TimeLens-8B. At 10%, however, the fixed rule raised both selectors by about 2.3 mIoU and 2.4 Acc@GQA, so we do not interpret the 5% differences until the reruns finish. SemVID at a nominal 5% is also not budget-matched to Coverage at 5%, because it keeps about three times as many visual tokens. **[TODO: 5% Coverage, random and global top-k with the fixed timestamp rule, both models; TimeLens-8B SemVID and 30%.]** We use the two-query protocol. Asking for the answer and the interval in one reply, as in prior zero-shot reports [16], lowers Acc@GQA by 16.5 for Qwen3-VL-8B and 9.2 for TimeLens-8B. The drop comes mostly from the time values, which the model then writes as minutes:seconds or copies from the prompt's example. Absolute numbers are therefore comparable only within a protocol.

Table 11. **NExT-GQA checks** (Qwen3-VL-8B, multiple choice, the 269 questions of the first 50 test videos; 10% rows use the fixed timestamp rule). The three rows below all tokens remove or scramble the video and are not ranked. SemVID 3.4% and 24.9% keep as many visual tokens as Coverage at 10% and 30%. The 10% block is ranked; the 30% rows are references for the paired comparison in the text.

Memory	Acc@QA	Acc@GQA	mIoP	mIoU
All tokens	83.6	40.1	49.0	37.6
No video	51.3	8.2	17.7	7.2
Frames shuffled	83.6	32.3	38.7	30.9
Timestamps shuffled	84.0	27.5	33.3	26.6
Random 10%	80.3	37.2	46.4	34.8
SemVID 10% (nominal)	78.8	35.3	44.7	32.4
SemVID 3.4% (matched)	71.4	31.2	41.4	28.9
Coverage 10%	**84.4**	**40.5**	**49.0**	**37.5**
SemVID 30% (nominal)	81.8	39.0	47.7	36.4
SemVID 24.9% (matched)	81.0	37.5	45.6	34.3

Checks and a matched-budget comparison. Table 11 removes or scrambles the video on a subset of NExT-GQA. Without video, accuracy falls by 32.3 points and grounding drops to 7.2 mIoU; the options alone still give half of the answers. Shuffling the frames leaves accuracy unchanged and lowers mIoU by 6.7. Shuffling only the timestamps also leaves accuracy unchanged (+0.4), but lowers Acc@GQA by 12.6 and mIoU by 10.9. The model reads the interval from the timestamps, but it does not need them for the answer. Keeping 100% of the tokens through either selector reproduces all 538 unpruned replies exactly. Batch size 1 instead of 8 leaves 502 of 538 replies unchanged and changes no metric by more than 0.4.

At the same number of visual tokens, Coverage leads SemVID on every metric. At 3.4% the margins are +13.0

Table 12. **NExT-GQA, open-ended** (Qwen3-VL-8B, 1,005 questions, options removed). Answers are judged correct when they match the ground truth in meaning (LLM judge, Video-ChatGPT prompt [36]); Acc@GQA also requires IoP ≥ 0.5. Two-query: the answer prompt plus the TimeLens grounding query. Joint: one answer with the interval, with the official S-EMBER prompt; joint intervals are scored with the official S-EMBER parser, which counts times written as minutes:seconds as missing. Rows come from separate judge batches of the same judge with a shared cache (a rerun of all tokens reproduces Acc@QA within 0.3). PCD uses the 64-token answer reserve. All 10% rows use the fixed timestamp rule (Tab. 10). All-tokens rows: batch size 2, first cluster; two-query 10% rows: batch size 8, second cluster; official-prompt rows: first cluster. § Rerun on the second cluster, whose own no-PCD rerun scores 33.2 / 17.1 / 48.0 / 37.5. Blocks are ranked separately.

Protocol	Method	Acc@QA	Acc@GQA	mIoP	mIoU
Two-query	All tokens	**33.5**	17.4	**48.0**	**37.7**
	All tokens + PCD§	33.1	16.6	45.9	36.1
	Random 10%	32.2	16.7	46.7	36.4
	SemVID 10%	29.1	13.7	42.5	31.0
	Coverage 10%	33.2	**17.7**	47.5	37.1
Joint, S-EMBER prompt	All tokens	38.7	13.2	35.1	30.4
	All tokens + PCD	**40.0**	13.4	33.4	28.9
	Random 10%	39.5	12.8	34.4	29.7
	Coverage 10%	38.9	**13.5**	**35.9**	**30.6**
	Coverage 10% + PCD	39.1	13.2	33.6	28.2

Acc@QA, +9.3 Acc@GQA, +7.7 mIoP and +8.6 mIoU. At 24.9% they are +4.1, +5.2, +3.6 and +3.1, where the Coverage 30% numbers come from the full run restricted to these questions. The fixed timestamp rule matters only for low-budget questions. On the 80 questions whose 10% budget is about 281 tokens, the earlier rule left Coverage at 2.4 mIoU, with 85% of its intervals starting at 0 s. The fixed rule gives 32.4.

Open-ended answers. Without the options (Tab. 12), answering becomes far harder (33.5 *vs*. 79.7 Acc@QA), while grounding is unchanged because the grounding query is the same. Our judge accepts the ground-truth answer for all 1,005 questions and agrees with letter correctness on 99.2% of the multiple-choice answers. Without pruning, Qwen3-VL-8B reaches 17.4 Acc@GQA zero-shot. TOGA [23], which is trained for open-ended grounded QA and judged with GPT-3.5, reaches 21.4. Coverage at 10% keeps the answers in both protocols, at −0.3 Acc@QA against all tokens in the two-query protocol (a cross-cluster pair) and +0.2 with the official S-EMBER prompt. In the two-query protocol random selection is close as well (Coverage +1.0 Acc@QA and Acc@GQA), and only SemVID is clearly worse. Coverage leads it by +4.2 Acc@QA, +4.0 Acc@GQA and +6.2 mIoU. In this protocol PCD slightly lowers grounding (−1.5 mIoU on all tokens, and −1.2 on Coverage in a run with the earlier timestamp rule) and barely changes Acc@GQA (−0.5 and −0.4). This matches what we saw with the one-line prompt on QVHighlights.

The official S-EMBER prompt on NExT-GQA. The official S-EMBER prompt asks for the answer and the interval in one reply, without a list of observations. Under this prompt, Coverage and random selection at 10% are both level with all tokens (+0.3 and −0.6 mIoU). PCD raises accuracy slightly on all tokens (+1.3) but lowers mIoU (−1.5 on all tokens, −2.5 on Coverage). Most of this loss is a format effect. PCD more than doubles the number of answers that write times as minutes:seconds, from 22 to 58 on all tokens and from 24 to 89 on Coverage, and the official parser scores such times as missing. In every run under this prompt, 68–76% of the intervals still start at 0 s.

TimeLens-8B, open-ended. For TimeLens-8B we have grounding results only: the answers are not yet judged, and the 10% runs use the earlier timestamp rule. In a joint answer with the timeline prompt, only 22–30% of its intervals start at 0 s. Here PCD lowers grounding, by −4.4 mIoU on all tokens, −4.5 on Coverage and −1.5 on random tokens. The replies show why. TimeLens-8B often ends with an answer and no `Time:` line, and PCD adds run-on or truncated replies. Replies without a parsable interval, which are scored as IoU 0, therefore rise from 37 to 112. In the two-query protocol, Coverage and random tokens at 10% are level (+0.1 mIoU), and both lose about 2 mIoU against all tokens. **[TODO: judge the TimeLens-8B open-ended answers; fixed-rule reruns of the PCD 10% rows.]**

On NExT-GQA, then, the contribution of the method is question-driven token selection. PCD does not raise Acc@GQA in either protocol of Tab. 12. Where it disturbs the answer format, as with TimeLens-8B or with minutes:seconds under the official prompt, it lowers the scored grounding.

6.3. QA pruners on grounding

Every published pruner (Fig. 4) falls below random token selection on three of the four datasets. On S-EMBER the losses range from −2.3 to −7.6 mIoU, on QVHighlights from −2.4 to −7.7, and on Charades-STA from −2.8 to −5.2. On ActivityNet at 10%, the input-level pruners are above random, which we discuss in Sec. 6.7. These gaps are not seed noise, because random selection varies little across seeds (26.0 ± 0.6 mIoU over three seeds on S-EMBER). The ordering follows coverage (Tab. 1). Methods that empty frames most aggressively lose the most, and those that spread tokens lose the least. Accuracy, by contrast, barely moves (Tab. 4). FastVID's accuracy is 2.1 points below random selection, about the decoding noise of our setup, while its mIoU is 7.6 lower. These pruners were designed to preserve answers, and they do, but they discard the timing. PCD helps every pruner (+7 to +12 mIoU) but does not close the gap to BOCC.

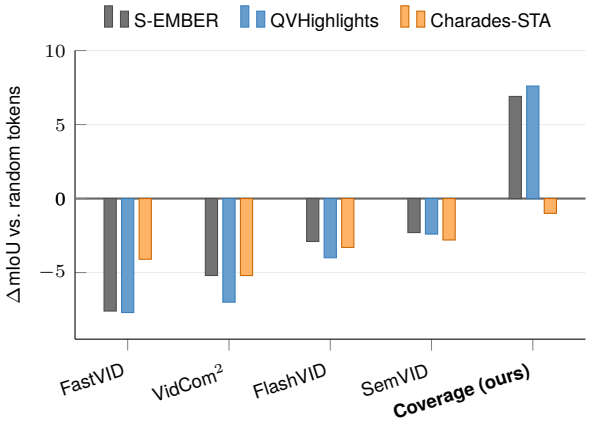


Figure 4. **QA pruners fall below random selection on grounding.** Paired mIoU difference to random token selection at the same 10% budget, greedy decoding. All four published pruners lose on all three datasets. Coverage gains on the two long-video datasets and is 1.0 mIoU below random selection on 30 s Charades videos.

Table 13. **Component ladder** (mIoU / R@0.5), timeline prompt, 10% of the tokens (ActivityNet: 30%); QVHighlights and ActivityNet: 500-query subsets. PCD: prior-contrastive decoding.

Configuration	S-EMBER	QVHighlights	ActivityNet
Uniform, random tokens	25.6 / 24.6	26.9 / 25.0	22.9 / 18.6
BOLT, random tokens	29.4 / 29.5	28.7 / 28.0	–
Coverage (uniform frames)	31.7 / 32.4	33.7 / 33.0	24.6 / 21.6
BOLT + Coverage	32.5 / 34.1	34.5 / 35.6	25.4 / 21.2
BOLT + Coverage + PCD	**39.3 / 40.8**	**48.8 / 54.8**	**34.7 / 32.4**

6.4. Analysis of Coverage

Contribution. Table 13 adds the components one at a time. On the memory side, Coverage gives the largest gain over random tokens, +6.1 mIoU on S-EMBER and +6.8 on QVHighlights. BOLT adds only +0.8 to +0.9 on top. Once tokens are chosen by the question, the choice of encoded frames matters less. **[TODO: Coverage + PCD without BOLT on all datasets.]**

Design choices. Table 14 varies one choice at a time. If there is a quota, an equal one works as well as any weighted one. We tried five relevance-weighted variants, using SigLIP similarity, attention mass or both, with floors of 25–50%. All are within ±0.5 mIoU of the equal quota (the uniform-frame SigLIP variant was our pre-registered primary test). Quotas that follow raw attention mass get pulled by attention sinks and recency bias, so that a single frame received up to 227 of about 2,140 kept tokens. The score itself should come from the actual question. HERMES's generic probe questions give 23.7 even with an equal quota, and 18.3 without one. Finally, deterministic top-k beats sampling from a coverage-floored density (31.7 *vs.* 30.0). A contrastive token score that subtracts the attention of a generic sentence, meant to cancel sinks, does not help.

Table 14. **Design choices of Coverage** (S-EMBER, greedy decoding). Frames: U = uniform. Δ *vs.* the equal quota with question top-k on the same frames.

Token selection	Frames	mIoU	Δ
Equal quota, question top-k	BOLT	32.5	–
Quota ∝ SigLIP relevance (floor 50%)	BOLT	**33.0**	+0.5
Quota ∝ SigLIP (floor 25%)	BOLT	32.6	+0.1
Quota ∝ SigLIP + attention	BOLT	32.9	+0.4
Quota ∝ attention mass	BOLT	32.1	−0.4
One token set shared by all layers	BOLT	31.9	−0.6
Equal quota, question top-k	U	31.7	–
Quota ∝ SigLIP relevance	U	31.4	−0.2
Equal quota, sampling instead of top-k	U	30.0	−1.7

Table 15. **Per-layer *vs*. shared token sets** (S-EMBER, 10%). Δ = shared − per layer.

Method	Per layer	Shared	ΔmIoU
Random tokens	25.6	26.3	+0.7
Random + PCD	35.0	35.8	+0.7
BOLT + Coverage	32.5	31.9	−0.6
BOCC	**39.3**	**37.8**	−1.5

Is the quota needed? For the generic HERMES score, the equal quota is worth +5.3 mIoU, because that score concentrates on the end of the window and empties earlier frames (Tab. 1). With the actual question's attention, the quota was not needed in either of the two settings we tested. There, a global top-k of the same per-layer attention, with no quota, matches the equal quota. On S-EMBER under the official prompt the difference is −0.1 mIoU (Tab. 20). On QVHighlights with BOLT frames and the one-line prompt it is −0.0 on the 584 queries run so far, where the global top-k is +1.8 above random tokens. In these settings, the gain of Coverage over random selection comes from scoring tokens by the question. The equal quota acts as a guarantee that every anchor survives, which matters for scores that collapse onto a few frames. **[TODO: global top-k without quota under the timeline prompt on S-EMBER and on the complete QVHighlights set (queued); share of frames left empty by it.]**

Per-layer selection. Table 15 compares one token set per layer with a single set shared by all layers. Sharing does not change random selection, so per-layer diversity is not why random selection beats input-level pruners. For BOCC, sharing costs −1.5 mIoU. Different layers attend to different tokens, and question-driven selection benefits from letting each layer choose its own.

6.5. Analysis of PCD

PCD on other memories. With reasoning-style answers, PCD can be added to any memory. Table 16 applies it to eight: three random budgets, the uncompressed model, HERMES and stratified pruning, and two HERMES streaming memories. PCD improves mIoU on all eight ($+4.6$ to $+8.0$) and GQ@0.5 on all eight ($+1.9$ to $+5.1$). It roughly halves the early starts everywhere, and its gains are largest where the memory is smallest. With 10% of the tokens and random selection, it already beats the uncompressed model (32.6 *vs*. 28.0). PCD also transfers to FastVID, the pruner that loses the most coverage, where it adds $+7.6$ mIoU and cuts the leak from 38% $\rightarrow$ 12%.

Choice of counterfactual. Table 17 changes only what the memory is contrasted against. The no-video input works best because it isolates the prior, and contrasting against it removes the prior (early starts 38% $\rightarrow$ 19%). Noised frames, as in VCD, still show part of the video *and* still produce the prior. The two runs then disagree mainly about content, and little of the prior is removed ($+0.5$ mIoU). Permuted timestamps also leave the prior in place. A one-sided rule that only penalises tokens the video argues against is worse than PCD (-5.6 and -4.8). Without video the model is about 99% sure of "0", so even a well-supported "0" looks like evidence against. Boosting visual attention during decoding, or stating the bias in the instruction, does not remove the prior either.

Time scope and adaptive strength. Contrasting every token also changes words that have nothing to do with time. The answer text changed in 85% of questions, and lists of more than eight moments rose from 12 to 56. Over-counting rose from 14% to 34% of counting questions. Restricting PCD to time values keeps the grounding gain and disturbs the text less (Tab. 18, top). PCD also risks hurting questions whose evidence really starts at 0 s. With a constant strength, their accuracy on the uncompressed memory falls from 21.8 to 16.7. The adaptive strength fades when the model is confident, and with it accuracy rises to 23.1 instead (Tab. 18, bottom; $+6.4$ *vs*. constant). The adaptive rule also gives the best *discrimination*, defined as early starts on early-evidence questions minus early starts on the others. On the all-tokens memory it reaches 56.4, against 49.6 for greedy decoding.

Other decision rules. We logged both distributions at the first time value of every answer and replayed other decision rules offline (Tab. 19). Even a threshold tuned on the gold labels cannot beat PCD's trade-off between keeping true early starts and removing leaked ones, so PCD is close to what decoding alone can do. In the early-start errors that remain, the memory itself is unsure, and fixing them would take more evidence in the memory.

Prompt format. With the official direct-answer prompt of S-EMBER (Tab. 20), BOCC still grounds better than the uncompressed model: $+5.5$ mIoU and $+5.9$ R@0.5 with the minutes:seconds-aware parser. Its accuracy gain is smaller, at $+2.9$. Most of the answer gain comes from BOLT. BOLT + Coverage is $+3.4$ above the uncompressed model in accuracy, and adding PCD to it changes accuracy by -0.4 while it raises mIoU by $+2.8$. So under this prompt PCD still moves the interval toward the evidence, but it no longer changes the answers. The official parser hides part of the interval gain. PCD makes the model write more times as minutes:seconds, and 73 BOCC answers cannot be read by the official parser, against 13–23 for the greedy rows. With that parser, BOCC is $+2.1$ above the uncompressed model and PCD changes mIoU by -0.8. The same format effect lowers the scored grounding of PCD on NExT-GQA under this prompt (-1.5 and -2.5, Sec. 6.2). Among the selectors, Coverage is level with global question top-k without a quota ($+0.1$ mIoU, -0.2 accuracy) and with SnapKV-style eviction (-0.8, $+0.6$), and it beats random tokens in accuracy ($+2.5$) and GQ@0.5 ($+1.9$). Random selection answers 2.5 points below the uncompressed model, while every other pruned memory is at most 0.6 points below it. These comparisons with the uncompressed model also carry the batch-size difference noted in the caption. Across prompts, BOCC with the official prompt is 4.7 mIoU below BOCC with the timeline prompt. **[TODO: PCD with and without BOLT under this prompt; the batch-matched timeline-prompt checks and the held-out split (queued).]**

We also tried an *event* timeline that lists start and end per occurrence (`Event: 12.0 -- 15.5 seconds`). Because it states extents, it improves the greedy baseline by $+7.4$ mIoU. PCD still adds $+2.1$ to $+3.3$ on top, reaching 35.0 mIoU and 12.6 GQ@0.5 at 768 tokens. The two fixes overlap. The format corrects extents, and PCD corrects the leak. BOCC uses the simpler timeline in all main results. **[TODO: event timeline with BOCC.]**

When does PCD help? The results so far suggest a simple rule. PCD helps in proportion to how much the video-free prior decides the time that is written. Its gain is largest where the memory says least, $+8.0$ mIoU for random 5% against $+4.6$ for the uncompressed model (Tab. 16). The adaptive strength fades when the memory makes the model confident, and so protects true early starts that a constant strength removes (Tab. 18). With a one-line answer, where the model states a single interval instead of listing observations, PCD has no effect (Sec. 6.2; those runs had no answer reserve). One qualification is needed, because PCD can also change the *form* of the answer. On NExT-GQA it lowers TimeLens-8B's grounding in the timeline format (-4.4 mIoU) by producing more replies without a parsable interval. Under the official prompt it makes Qwen3-VL-8B write more times as minutes:seconds (Sec. 6.2). In short, PCD helps when the prior chooses the times and the answer keeps its format. We state this as a hypothesis, which the

Table 16. **PCD on eight memories** (S-EMBER, Acc. / mIoU / GQ@0.5; earlier implementation of PCD). Plain PCD: on all tokens with constant α=0.5; PCD: confidence-adaptive strength on time values only (ours). Δ PCD *vs.* greedy. Best and second best per row.

Memory	Greedy	Plain PCD	PCD	ΔmIoU	ΔGQ@0.5	Early starts
Random 5%	15.8 / 23.6 / 5.9	15.4 / 29.2 / 9.1	**16.4 / 31.5 / 9.9**	+8.0	+4.0	39% → 20%
Random 10%	15.2 / 25.8 / 7.2	**16.2** / 32.1 / **10.1**	**16.2 / 32.6 / 10.1**	+6.8	+2.9	38% → 19%
Random 25%	17.7 / 28.2 / 8.2	**18.9** / 33.0 / 11.8	**18.9 / 34.9 / 13.3**	+6.7	+5.1	37% → 21%
All tokens	18.5 / 28.0 / 8.6	19.8 / 31.5 / 11.2	**21.5 / 32.7 / 13.3**	+4.6	+4.6	41% → 23%
HERMES 10%	13.1 / 18.3 / 4.0	**15.4** / 22.6 / **7.2**	13.3 / **24.1** / 5.9	+5.7	+1.9	56% → 31%
Stratified 10%	14.1 / 23.7 / 6.1	14.5 / 29.9 / 8.8	**15.8 / 31.0 / 10.5**	+7.3	+4.4	36% → 17%
HERMES streaming 4k	13.5 / 21.6 / 4.0	**17.1** / 27.4 / **8.2**	15.4 / **28.6** / **8.2**	+6.9	+4.2	45% → 24%
HERMES streaming 6k	14.3 / 22.6 / 4.8	14.7 / 28.1 / 7.6	**15.4 / 29.8 / 8.4**	+7.2	+3.6	46% → 24%
Mean of 8	15.3 / 24.0 / 6.1	16.5 / 29.2 / 9.3	**16.6 / 30.7 / 10.0**	+6.7	+3.9	

Table 17. **What to contrast against** (S-EMBER, random 10%).

Counterfactual	mIoU	GQ@0.5	Early
None (greedy)	25.8	7.2	38%
No video (ours)	**32.1** (+6.2)	**10.1** (+2.9)	**19%**
Noised frames (VCD [30])	26.3 (+0.5)	7.8 (+0.6)	30%
Evidence-against rule, α=1 / 2	26.5 / 27.3	–	–
Attention gain (PAI-style)	+1.3	+0.8	42%
Instruction "do not assume early"	25.7	6.9	37%

Table 18. **Time scope and adaptive strength** (S-EMBER, Acc. / mIoU / GQ@0.5). Top: PCD on all tokens *vs.* time values only. Bottom: the 78 questions whose evidence truly starts in the first 5% of the window. Top: best per row.

Memory	All tokens	Time values only
Random 10%	**16.2** / 32.1 / 10.1	16.0 / 32.1 / **10.3**
Random 25%	**18.9** / 33.0 / 11.8	17.7 / **34.0** / 11.8
All tokens	19.8 / 31.5 / 11.2	**20.8 / 32.9 / 12.4**
HERMES 10%	**15.4** / 22.6 / 7.2	14.7 / **24.3** / **7.4**
Streaming 4k	**17.1** / 27.4 / 8.2	16.2 / **29.2** / **9.3**
Early-evidence questions (78), all tokens memory		
Greedy	21.8 / 49.2 / 16.7	
PCD, constant α	16.7 / 55.3 / 12.8	
PCD, adaptive α	**23.1 / 56.9 / 19.2**	

Table 19. **Decision rules replayed at the first time value** (S-EMBER, random 10%). Share of answers starting at "0".

Rule	Early evidence ↑	Other (leak) ↓
Greedy	**69%**	16%
PCD, α=0.5	46%	**6%**
PCD, adaptive	51%	7%
Evidence-against, α=4	51%	9%
Skewed or tempered prior	50–55%	8–11%
Best threshold (uses gold)	46%	**6%**

Table 20. **Official S-EMBER prompt** (direct answer with `Answer:` and `Time:` lines; Qwen3-VL-8B, 64 frames, 10%, second cluster; judge session C). mIoU / R@0.5 with the minutes:seconds-aware parser; mIoU with the official parser and the number of answers it cannot parse. The uncompressed model ran at batch size 4 (8 ran out of memory), the pruned rows at 8.

Method	mIoU / R@0.5	Acc. / GQ@0.5	Official mIoU	Unparsed
All tokens	29.1 / 28.4	13.7 / 4.8	27.4	20
Random tokens	27.9 / 26.1	11.2 / 3.6	26.4	17
Coverage, uniform frames	30.1 / 29.3	13.7 / 5.5	28.8	**13**
Global question top-k, no quota	30.0 / 29.3	13.9 / 5.1	28.5	17
SnapKV-style eviction	30.9 / 29.5	13.1 / 5.1	29.6	16
BOLT + Coverage	31.9 / 32.2	**17.1** / 5.9	**30.2**	23
BOCC	**34.6 / 34.3**	16.6 / **6.3**	29.5	73

per-token log-probabilities we already store can test directly. **[TODO: PCD gain against the divergence between $p_{\mathcal{M}}$ and $p_{\varnothing}$ at each written time value, across all conditions; one prediction stated before the run.]**

6.6. Budget

Table 21 and Fig. 5 vary the budget at 64 frames and show two regimes. Above roughly 500 kept tokens, question-driven selection stays flat or even improves as the budget *shrinks*. On S-EMBER, BOLT + Coverage rises from 31.0 at 20% to 34.6 at 5%, and BOCC from 36.8 to 39.3. Random selection and SemVID degrade over the same range. Pruning by the question thus removes distractors as well as cost. At 5%, BOCC leads SemVID + PCD by +7.4 mIoU, and at 20% by +1.5.

Below roughly 300 kept tokens (QVHighlights 5%, ActivityNet 10%), the equal quota leaves only 2–4 tokens per frame pair. The top-attended tokens then carry little content, and BOCC collapses to 9.7 and 11.6. This is below random selection and below SemVID, whose cross-frame merging degrades more gracefully. What matters is the absolute number of tokens per frame, which is low in our re-encoded QVHighlights and ActivityNet copies. One caveat applies. These runs count the kept timestamps inside the

Table 21. **Budget sweep at 64 frames** (mIoU). The second row gives the number of kept tokens per memory.

	S-EMBER			QVHighlights			ActivityNet		
Budget	5%	10%	20%	5%	10%	20%	10%	30%	50%
Kept tokens	∼1,070	∼2,140	∼4,280	∼260	∼520	∼1,040	120–300	400–900	600–1,500
Random	23.9	25.6	27.4		26.9		17.8	22.9	23.5
SemVID	20.9	23.3	26.5		24.5		21.2	22.8	22.3
R + C (no PCD)	34.6	32.5	31.0		34.5		11.3	25.4	25.0
SemVID + PCD	31.9	34.8	35.3		34.8		**26.8**	30.2	31.4
Random + PCD	34.5	35.0	36.5		37.8		21.5	31.4	32.2
BOCC	**39.3**	**39.3**	**36.8**	9.7	**48.8**	49.4	11.6	**34.7**	**34.4**

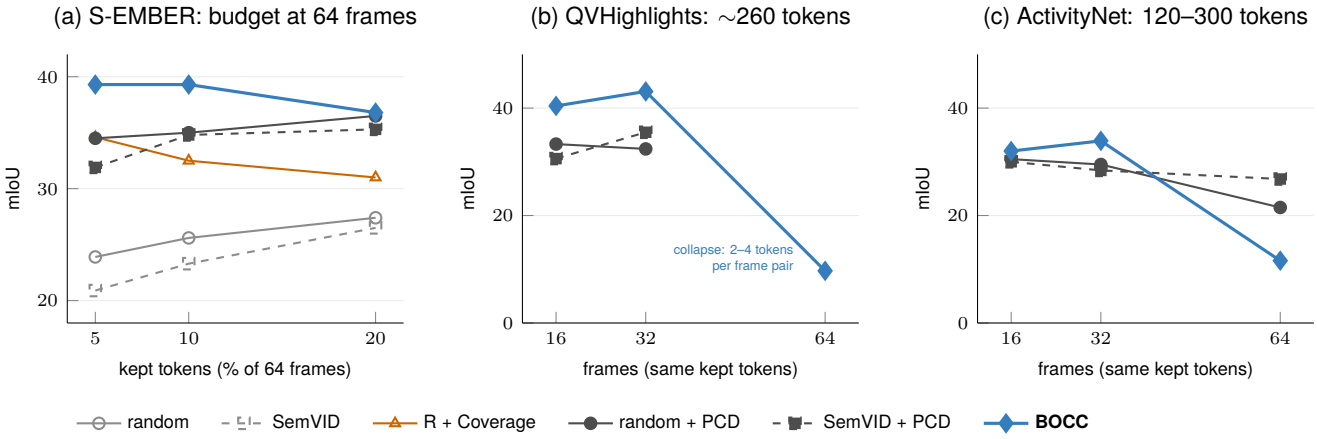


Figure 5. **Budget and how to spend it. (a)** At 64 frames, question-driven selection improves as the budget shrinks while random selection and SemVID degrade; BOCC leads at every budget that leaves at least about eight tokens per frame pair, and collapses below it (QVHighlights 5%, ActivityNet 10%; these runs use the earlier timestamp rule, Sec. 6.6). **(b, c)** At a fixed, very small number of kept tokens, 64 frames leave 2–4 tokens per frame pair and BOCC collapses; spending the same tokens on 32 frames recovers it (QVHighlights 9.7 → 43.1, ActivityNet 11.6 → 33.9).

budget under the earlier timestamp rule, which on NExT-GQA left some low-budget questions (about 280 tokens) with no visual tokens (Sec. 6.2). Part of the collapse below about 300 tokens may therefore come from this accounting rather than from the equal quota. **[TODO: QVHighlights 5% and 10% at 64 frames with the fixed timestamp rule (queued).]**

6.7. Fewer frames or fewer tokens per frame?

The collapse at small budgets comes from how the budget is spent, and the equal quota is not at fault in itself. The timestamp-accounting caveat of Sec. 6.6 still applies here, since fewer frames also means fewer timestamps inside the budget. In Tab. 22, every row of a block keeps the same number of tokens, and only the split between frames and tokens per frame changes. With 32 frames instead of 64, each frame pair keeps twice as many tokens, and BOCC recovers. On QVHighlights it goes from 9.7 to 43.1 mIoU. On ActivityNet it goes from 11.6 to 33.9, which beats SemVID + PCD by +5.5 and nearly matches its own 64-frame result at three times the budget (34.7). Halving again to 16 frames loses a little, because past a point temporal coverage becomes the bottleneck. The baselines also gain from fewer frames, but less. On Charades-STA, whose videos last 30 s, fewer frames do not help any method. We therefore recommend the rule of Sec. 4.1. Keep at least about eight tokens per frame pair, and reduce F when the budget does not allow it. At equal tokens, more frames with fewer tokens each do not help either. With 128 frames at 5% on S-EMBER, the change is +1.2 mIoU without PCD and −3.6 with it, with 74–78 unparseable answers.

6.8. Comparison with gold-informed upper bounds

Table 23 gives the memory information it should not have. On Qwen3-VL-8B, BOCC beats perfect token selection at the same budget without using any gold information. Over the oracle selector with PCD, it gains +4.4 mIoU and +4.0 GQ@0.5. Knowing which tokens to keep does not stop the prior, and keeping only the gold frames' tokens starves the rest of the timeline. The strongest oracle sees only the evi-

Table 22. **Spending a fixed token budget on fewer frames** (mIoU / R@0.5, all with PCD). Within each block every row keeps the same number of tokens: F frames at keep ratio $\rho \cdot 64/F$. Best and second best per row.

Dataset (kept tokens)	Frames × keep ratio	Random + PCD	SemVID + PCD	BOCC
QVHighlights (∼260)	64 × 5%	–	–	9.7 / 6.6
	32 × 10%	32.4 / 29.2	35.5 / 33.6	**43.1 / 46.6**
	16 × 20%	33.3 / 32.4	30.6 / 28.0	**40.4 / 41.0**
ActivityNet (120–300)	64 × 10%	21.5 / 18.2	**26.8 / 23.6**	11.6 / 7.6
	32 × 20%	29.5 / 26.8	28.4 / 25.8	**33.9 / 32.0**
	16 × 40%	30.5 / 26.8	30.0 / 27.6	**32.0 / 29.8**
Charades-STA (∼10% of 64)	64 × 10%	**26.2 / 24.0**	21.7 / 19.2	23.3 / 21.2
	32 × 20%	**24.6 / 22.0**	20.8 / 16.0	23.6 / 21.0[§]

[§] Coverage on uniform frames; BOLT is skipped on Charades-STA.

Table 23. **Upper bounds that use the gold interval** (S-EMBER, Qwen3-VL-8B, 475 questions; Acc. and GQ@0.5 from judging session B; minutes:seconds-aware parser). Oracle selector: the gold frames' tokens are kept first. Gold window: all 64 frames inside the gold interval (batch size 3). Leak: intervals starting in the first 5% of the window, on the 397 questions whose evidence starts later.

Memory	Gold	Acc.	mIoU	R@0.5	GQ@0.5	Leak
Random 10%	–	17.1	25.6	24.6	7.2	29%
Random 10% + PCD	–	20.0	35.0	33.9	12.0	14%
Oracle selector 10%	✓	18.9	27.9	25.7	9.3	33%
Oracle selector 10% + PCD	✓	18.5	34.9	35.2	9.5	16%
BOCC 10%	–	21.9	39.3	40.8	13.5	**11%**
All tokens	–	19.6	28.0	26.7	8.8	33%
Gold window	✓	**29.5**	36.1	38.1	**15.8**	45%
Gold window + PCD	✓	27.2	**41.5**	**43.2**	14.9	35%

Table 24. **Gold-informed upper bounds for TimeLens-8B** (S-EMBER, 475 questions; all Acc. and GQ@0.5 from judging session C, Sec. 5). Settings as in Tab. 23; all tokens at batch size 1. No oracle selector without PCD was run.

Memory	Gold	Acc.	mIoU	R@0.5	GQ@0.5
Random 10%	–	18.5	28.2	26.7	9.1
Random 10% + PCD	–	19.8	38.4	39.4	11.8
Oracle selector 10% + PCD	✓	21.7	43.3	45.3	14.3
BOCC 10%	–	26.5	43.9	46.9	17.1
All tokens	–	19.8	33.1	34.1	11.6
Gold window	✓	28.2	46.6	46.7	19.2
Gold window + PCD	✓	**36.2**	**54.9**	**57.1**	**25.9**

dence and also uses PCD. Against it, BOCC is lower by 2.2 mIoU and 1.5 GQ@0.5, and it answers worse (−5.3 accuracy). Even the gold window leaks 45%, so with every frame inside the evidence the model still often starts near 0 s. On the gold window, PCD lowers the leak to 35% and adds +5.5 mIoU, while accuracy changes by −2.3. For Qwen3-VL-8B, then, more evidence raises accuracy, while PCD fixes the time values.

For TimeLens-8B (Tab. 24), the bounds are much higher and BOCC is further from them. Over the oracle selector with PCD, BOCC gains +0.5 mIoU, +2.7 GQ@0.5 and +4.8 accuracy. The gold window alone is +13.2 mIoU above all tokens (both at batch size 1). Adding PCD on the gold window raises mIoU by +8.3, and it also raises accuracy (+8.0) and GQ@0.5 (+6.7). BOCC stays below this bound by 11.0 mIoU, 9.7 accuracy and 8.8 GQ@0.5. So for the grounding-tuned model, PCD and evidence work together. With the right frames, contrasting the time values also improves the answers. The gap to the gold window shows how much a better choice of frames could still add for this model.

6.9. Analysis of BOLT

The SigLIP 2 index does find the evidence. Over the seconds before each question, the AUROC of evidence seconds against the rest is 0.663. The top-ranked second lies in the gold interval for 63% of questions (chance 36%), and for 41% of location questions (chance 13.5%). BOLT frames put 40% of the memory inside the evidence, against 36% for uniform frames. This helps random tokens (+3.8 mIoU). Once tokens are chosen by the question, however, BOLT adds only +0.8 to +0.9, because both stages focus the memory on the question. Subtracting a question-free prior from SigLIP's score does not help (AUROC 0.635–0.666). This is expected for a retriever trained with a contrastive loss, whose score already estimates pointwise mutual information [45]. Re-weighting the token budget of uniform frames by the same relevance adds +1.3. The choice of encoded frames matters more than how the budget is split among fixed frames.

6.10. What did not work

Table 25 collects the ideas that gave little or nothing. Taken together, they suggest that at a fixed budget, redistributing

Table 25. **Negative results** (S-EMBER; ΔmIoU against the stated reference).

Idea	Reference	ΔmIoU
VCD-style noised-frame contrast	greedy	+0.5
Permuted-timestamp contrast	greedy	+0.9
Evidence-against decoding rule	PCD	−5.6
Attention gain while decoding	greedy	+1.3
Bias stated in the instruction	greedy	−0.1
Relevance-guided frame zoom	greedy	−1.5
Zoom inside the gold window	greedy	+2.3
Unequal per-frame quotas (5 variants)	equal quota	−0.4 to +0.5
Contrastive token score	question top-k	−1.2 to +0.5
Prior-subtracted retrieval	SigLIP	AUROC $\leq$ +0.003
128 frames at the same budget	64 frames	+1.2 / −3.6

Table 26. **Error types** on S-EMBER, pooled over eight memories, greedy *vs*. PCD (earlier implementation of PCD).

Outcome	Greedy	PCD	Δ
IoU $\geq$ 0.5, answer correct	6.1%	9.9%	+3.8
IoU $\geq$ 0.5, answer wrong	16.3%	20.7%	+4.4
Partial overlap (IoU $<$ 0.5)	17.0%	15.7%	−1.3
Inside gold, too short	22.2%	18.5%	−3.7
Entirely *before* gold	33.1%	18.9%	−14.2
Entirely after gold	2.7%	5.7%	+3.0
Covers gold, too long	2.2%	2.5%	+0.3
No answer / unparsed	0.5%	8.1%	+7.6

evidence by zooming, re-weighting frames or unequal quotas is close to its ceiling. Large gains came only from removing the prior where it acts (PCD), or from selecting tokens by the actual question in every layer, with or without the equal quota (Sec. 6.4).

6.11. Where answers still fail

Table 26 shows that PCD works where it was aimed. Intervals entirely before the gold one are the signature of the prior. Under greedy decoding they are the largest error type, and with PCD they fall to less than a fifth of answers. The remaining failures point to four limits. (i) Wrong answers on good intervals. Twice as many answers are well grounded but wrong as are grounded and right. Counting answers do not use the events the model itself listed, and an IoU of 0.5 still allows a two-fold duration error. (ii) Recent events. Of the intervals that still end before the gold one, 81% have evidence in the second half of the window, and the model picks an earlier similar event instead. When the evidence lies in the last third of the window, GQ@0.5 is 4.4. (iii) Sparse evidence. With fewer than 4 of the 64 frames inside the evidence, GQ@0.5 is 1–2% for any method, against 15% with 16 or more. No decoding rule can recover evidence that is not in the input. (iv) Label ambiguity. On the 98 questions whose annotators disagree (mean pairwise IoU $<$ 0.6), every method scores near zero.

7. Limitations

Answer format. The S-EMBER results and most analysis tables use a reasoning-style timeline prompt. On single-moment grounding this prompt costs the uncompressed model about 30 mIoU compared with the standard one-line prompt (Tab. 6), so the gains of BOCC over the uncompressed model on our QVHighlights, ActivityNet and Charades-STA subsets (Tab. 13) hold only relative to this format. Under the one-line prompt, on the full Charades-STA and QVHighlights sets, we have so far run Coverage and random selection but not BOCC or the QA pruners (Tabs. 3 and 8), and we have not yet run the full ActivityNet set. On S-EMBER, BOCC beats the uncompressed model when both answer in the timeline format. Under the official direct-answer prompt it still grounds better (+5.5 mIoU with our parser, +2.1 with the official one, which cannot read 73 of its answers), but its accuracy gain is smaller (+2.9) and comes mostly from BOLT; the uncompressed model there ran at batch size 4 and the pruned rows at 8 (Tab. 20). **[TODO: S-EMBER without pruning under the one-line prompt; batch-matched comparison.]**

Where the prior is measured. All our measurements of the video-free prior (Sec. 3.3) use the timeline format. With a one-line answer, PCD has no effect (in runs without the answer reserve), which suggests that the prior decides few or none of the times written in that format. We have not yet measured the no-video answers in it. **[TODO: no-video answers under the one-line and direct-answer prompts.]**

What drives Coverage. The equal quota guarantees that every timestamp survives, and it helps scores that concentrate on a few frames, such as the HERMES probes (+5.3). With attention from the actual question, however, a global top-k without the quota matched it in both settings we tested, S-EMBER with the official prompt and a partial run on QVHighlights with BOLT frames. Our evidence that grounding needs anchors in every frame therefore comes from selectors that empty frames, namely the QA pruners and the generic probe scores, and does not rest on the quota itself. **[TODO: global top-k without quota under the timeline prompt; frame coverage of question top-k.]**

Minimum tokens per frame. Equal quotas fail when a frame pair keeps only 2–4 tokens. Spending the budget on fewer frames avoids this (Sec. 6.7), but we derived our rule of at least about eight tokens per frame pair from those same results, and it still needs validation on held-out data. The small-budget runs also use the earlier timestamp rule, which left some NExT-GQA questions with no visual tokens, so part of the collapse may be an accounting effect. **[TODO: small-budget runs with the fixed timestamp rule.]**

Short, densely sampled videos. On Charades-STA, uniform frames are less than a second apart, and on NExT-GQA 64 frames already cover a 40 s video. On such

videos question-driven selection improves grounding only slightly over random selection: by $+0.5$ and $+0.7$ mIoU on Charades-STA, and by $+0.6$ and $+1.1$ mIoU on NExT-GQA at 10%. Its benefit there shows up in the answers instead. On NExT-GQA at 10% it keeps the accuracy of the uncompressed model, while random selection loses about four points. The 5% NExT-GQA results use the earlier timestamp rule and are being rerun. On these videos PCD does not raise grounded accuracy in either protocol of Tab. 12, and it lowers the scored interval in both. **[TODO: 5% NExT-GQA with the fixed timestamp rule.]**

Where PCD hurts. PCD is meant to change which time values are written, but it can also change how they are written. On NExT-GQA it lowers TimeLens-8B's timeline grounding by -4.4 mIoU, because more replies end up without a parsable interval. Under the official prompt it makes Qwen3-VL-8B write more times as minutes:seconds, which the official parser scores as missing. In deployment one would need a parser that accepts both forms, or a format check on the contrasted reply.

Upper bounds and the front end. BOCC remains well below the gold-window upper bound of TimeLens-8B with PCD, by -11.0 mIoU and -9.7 accuracy. For Qwen3-VL-8B it is below the gold window in accuracy (-5.3). With BOLT frames, most of the gain over a low-resolution input comes from frame selection and resolution, and random tokens on the same frames get it too. We also do not count the cost of the SigLIP index.

Not a streaming memory. We rebuild the memory for each question from the frames that precede it. The memory is causal, but the full memory is prefilled once per question, so peak memory and time-to-first-token match those of the uncompressed model. BOCC reduces the memory the decoder reads and leaves the cost of building it unchanged. PCD also decodes a second, prompt-only row at every step. **[TODO: cost table; amortisation over several questions per video; write-once memory re-ranked at question time.]**

Scope of the evaluation. Both of our models come from one family. TimeLens-8B is fine-tuned from Qwen3-VL-8B, and both read time from text timestamps. Most diagnostic results use subsets of 475–500 questions. Answers are judged by a local model with the official prompt, in three judging sessions that differ by up to about one accuracy point. Some paired differences combine runs from two GPU clusters, whose repeated runs agree within one mIoU point. We chose every hyperparameter on the S-EMBER questions. The other datasets were run once with the frozen configuration, but we have not yet evaluated a held-out S-EMBER split. **[TODO: a second model family; held-out S-EMBER split.]**

8. Conclusion

Pruning for QA keeps what is salient, but grounding needs what is everywhere. A video LLM reads time from timestamp anchors that survive only together with their frames. Pruners built for QA empty the frames that look unremarkable, and as a result they fall below random selection on grounding. If each layer instead keeps the tokens that the actual question attends to, the timeline stays usable. With a quarter of the visual tokens, this matches the uncompressed model under the standard grounding prompt on the full Charades-STA and QVHighlights sets, and with a tenth of the tokens it keeps NExT-GQA accuracy, for both a general and a grounding-tuned model. An equal per-frame quota guarantees every anchor, although with the question's attention it was not needed in the settings we tested.

A second failure appears when the answer lists what was seen before it states the interval, as grounded QA does. Where the memory is silent, a video-free prior writes the times. Contrasting each time value with the model's own no-video prediction removes this prior, and with a tenth of the tokens BOCC then grounds better than the uncompressed model answering in the same format. Where the contrast disturbs the answer format, though, it can lower the scored grounding. The answer format itself changes single-moment grounding by about 30 mIoU. It should be reported in every evaluation of compressed video memories.

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A. Parsing times written as minutes:seconds

On videos longer than a minute, the model sometimes writes times as minutes:seconds, *e.g.* `Time: [1:45, 2:07]`. The official S-EMBER parser scores such times as missing, so we convert `h:mm:ss` and `m:ss` to seconds, for every method alike. With the official parser the method rankings do not change. The timeline-prompt numbers of BOCC in Tab. 13 become 44.6 / 50.2 on QVHighlights (*vs*. 48.8 / 54.8) and 33.3 / 31.0 on ActivityNet at 30% (*vs*. 34.7 / 32.4). **[TODO: full table with the official parser.]**

B. Checks against published numbers

Before comparing pruning methods under the TimeLens-Bench protocol, we reproduce the published unpruned scores with the TimeLens evaluation script. We also check that our own inference pipeline gives the same scores as the script on the same queries (Tab. 27).

Table 27. **Reproduction of TimeLens-Bench** (mIoU, no pruning). Top: TimeLens script on the full sets *vs*. published. Bottom: our pipeline *vs*. the TimeLens script on the same queries (paired Δ).

Model	Dataset (queries)	Ours	Reference	Δ
Qwen3-VL-8B	Charades-STA (3,363)	48.5	48.3 (published)	
Qwen3-VL-8B	QVHighlights (1,541)	59.6	59.4 (published)	
Qwen3-VL-8B	ActivityNet (4,500)	47.1	46.8 (published)	
TimeLens-8B	Charades-STA (3,363)	55.4	55.2 (published)	
TimeLens-8B	QVHighlights (1,541)	65.7	65.5 (published)	
TimeLens-8B	ActivityNet (4,500)	53.2	53.2 (published)	
Qwen3-VL-8B	Charades-STA (400)	48.0	47.5 (script)	+0.5
Qwen3-VL-8B	ActivityNet (400)	47.1	45.7 (script)	+1.4
Qwen3-VL-8B	QVHighlights (513)	60.5	60.5 (script)	0.0
TimeLens-8B	QVHighlights (513)	67.1	66.1 (script)	+1.0

C. Reproduction

BOCC corresponds to the following overrides of our code, which we release with the paper:

```
model=qwen3_vl_8b experiment=sember_grounding_uniform run.uniform_num_frames=64
dataset.grounding_prompt=timeline run.keep_time_tokens=surviving run.max_new_tokens=384
run.batch_size=8 run.prune_score=core_topk run.offline_keep_ratio=0.1
run.contrastive_mode=blind run.contrastive_alpha=1.0 run.contrastive_adaptive=true
run.contrastive_scope=time
(BOLT) run.retrieval=frames_cont run.retrieval_min_gap=1.0
```